

Reinforcement learning for firms in a large-scale macroeconomic ABM: emerging pricing strategies and forecasting performance

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Abstract

We extend a large, heterogeneous agent-based economic simulation model (ABM), calibrated to real data and representing the Italian macroeconomic system, through the introduction of a multi-agent reinforcement learning mechanism for the price-setting behavior of firms, replacing the simple heuristics typically used in ABMs. We find that firms learn a mean-reverting price-setting behavior, adjusting prices in response to deviations from the sectoral average, which is consistent with both empirical observations and theoretical models. We then apply this behavior to different shares of firms and evaluate the forecasting performance on macroeconomic variables. Our results show that modeling the behavior of firms with reinforcement learning can improve the model's fit to macroeconomic data, but only in specific settings and if not all firms are set to learn.

Keywords: Agent-based models - Behavioral macroeconomics - Calibration - Macroeconomic forecasting - Simulation

JEL: E70, E32, E37

1. Introduction

The 2008 Great Recession prompted theoretical debates on the limitations of mainstream macroeconomic models, leading to a demand for frameworks that account for heterogeneity, nonlinearities, domino effects, financial systemic risk, and credit crunches (Dosi, 2011; Kirman, 2010; Krugman, 2011; Stiglitz, 2011a). Another major limitation of mainstream models is the assumption of perfectly rational agents, contrasting the extensive evidence from the behavioral economics literature initiated by Simon (1955) and Kahneman and Tversky (1979).

Agent-based macroeconomic models (ABMs) have emerged as a viable alternative to neoclassical models due to their flexibility in capturing heterogeneity, interactions, and nonlinear dynamics. Moreover, ABMs use various heuristics to model agents' behaviors to better reflect the behavioral patterns observed in the behavioral economics literature. However, traditional ABMs have faced criticism for several main limitations. First, they primarily offer qualitative insights and are challenging to align with empirical data at the micro, meso, or macroeconomic level. In

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other words, they “only” qualitatively replicate stylized facts². Moreover, the heuristics used to model agents’ behaviors can sometimes be criticized for being arbitrary or for modeling agents as overly irrational. More broadly, ABMs suffer from an excess of degrees of freedom due to the wide range of possible behavioral rules and the large number of parameters, typically requiring extensive and time-consuming calibration procedures (Platt, 2020; Benedetti et al., 2022; Glielmo et al., 2023). These limitations have hindered their adoption as practical tools for policy analysis.

In response to criticisms regarding the provision of only qualitative analyses, excessive degrees of freedom and a large number of parameters requiring calibration, a new strand of ABMs has begun to emerge, starting with Poledna et al. (2023) and followed by Hommes and Poledna (2023), Hochrainer-Stigler et al. (2024), Poledna et al. (2024), Hommes et al. (2025) and Di Domenico et al. (2025). These models adopt a data-driven approach to parameter calibration and are better suited for conducting quantitative analyses.

To address the criticism of modeling agents as overly irrational, some recent studies have begun to incorporate machine learning (ML) techniques into agent behavior modeling. For example, ML has been applied in small theoretical macroeconomic ABMs such as: Catullo et al. (2022), which use genetic algorithms to model firms’ sales expectations; Curry et al. (2023), which employ multi-agent reinforcement learning (MARL) for firms, households, and government; and Brusatin et al. (2024), which apply MARL to firms’ price and quantity decisions. In this way, although not perfectly rational, agents display a higher level of intelligence as they attempt to optimize their behavior based on contextual information and past experience.

Unlike the aforementioned works, in this paper, we apply MARL to the price-setting behavior of firms within a large, data-driven macroeconomic model, thereby integrating the two aforementioned strands of research. Our starting point is the model developed by Poledna et al. (2023), calibrated to the Italian economy as in Di Domenico et al. (2025). Our objective is twofold: (i) to observe which pseudo-optimal pricing rules emerge with MARL when heterogeneous agents interact in a complex environment; (ii) to assess whether the inclusion of agents (in this case, firms) with a higher level of rationality compared to the use of simple heuristics can improve the out-of-sample forecasting of macroeconomic variables.

About the first research question, our main result is a mean-reverting price-setting rule: firms adjust their prices in response to deviations from the sectoral average, displaying a negative relationship between price changes and their distance from the mean. This behavior aligns with empirical evidence from advanced economies, where firms tend to adjust prices toward competitors’ average levels (Kim, 2025; Koga et al., 2020). Reinforcement learning thus enables the endogenous emergence of realistic, context-sensitive micro-behavior consistent with observed market patterns. The relationship becomes more nuanced when firm size influences the search and matching process. Smaller firms, particularly in highly populated sectors, reduce prices more aggressively to defend or increase market share, shifting the mean-reverting patterns slightly downward. In contrast, firms with large market shares in less crowded sectors exhibit more dispersed behavior, suggesting that some firms exploit their market power and depart from the mean-reverting dynamic.

Regarding the second research question, including a non-negligible fraction (25%–50%) of RL firms may improve the predictive performance of the model of Poledna et al. (2023). However, this occurs only under appropriate parameter settings, as in other cases, the presence of RL firms worsens the forecasts.

The paper is structured as follows. Section 2 summarizes some relevant related literature. Section 3 describes

²Anyone who builds ABMs knows that even qualitatively replicating several stylized facts requires a demanding calibration effort.

the methods adopted: Section 3.1 outlines the main features of the model of Poledna et al. (2023), while Section 3.2 presents the reinforcement learning algorithm used to guide firms’ pricing decisions. Section 4 reports the simulation results addressing the first research question and Section 5 the second. Section 6 concludes.

2. Literature review and use of MARL

In mainstream economics, agents are often modeled as *rational decision-makers* who can process all available information and make optimal choices to maximize utility or profit. These *rational agents* are assumed to have complete foresight, or at least rational expectations, and they adjust their behavior in response to market signals, prices, and incentives in an optimal fashion. This concept is central to much of *neoclassical economic theory*, where the *representative agent* is used to model aggregate behavior, representing an economy-wide decision-maker who acts rationally and consistently. Under the Rational Expectations (RE) assumption, a significant portion of the profession considers expectations uniform across agents (with some allowance for noise) and, on average, accurate. Consequently, actions have been assumed to align with those that optimize a given objective, based on accurate expectations about the future, which depend on all available information. However, these claims regarding expectations or actions are neither supported by empirical evidence (Carroll, 2003; Coibion and Gorodnichenko, 2012, 2015; Gennaioli et al., 2016) nor corroborated by experimental studies (Tversky and Kahneman, 1974; Schweitzer and Cachon, 2000; Kahneman, 2003; Anufriev and Hommes, 2012). Moreover, the very notion of “rational” expectations becomes unclear, particularly under conditions of Knightian uncertainty, significant policy shifts (Stiglitz, 2011b, 2015), or structural breaks in the underlying distributions on which agents base their forecasts (Hendry and Mizon, 2010).

In contrast, *bounded agents*, often used in *behavioral economics* and *agent-based modeling*, operate under bounded rationality, making decisions based on heuristics, limited information, or simplified rules of thumb. These agents are not constantly optimizing; instead, they rely on predefined decision rules, and sometimes adapt to their environment, switching their decision rules, in ways that more closely resemble real-world human behavior. Using bounded agents allows for a more realistic representation of economic dynamics, as it captures the complexity and heterogeneity of economic decision-making. While analytically convenient, the *representative agent* assumption in traditional models often overlooks this complexity. It fails to account for systemic risks or emergent aggregate behaviors arising from interactions among heterogeneous agents (Kirman, 1992).

Importantly, recent work show that bounded rationality may not simply be a pragmatic simplification but can outperform more sophisticated decision rules. Dosi et al. (2020) demonstrate that, in complex macroeconomic environments, agents who rely on simple heuristics may achieve higher prediction accuracy and welfare outcomes than those adopting more elaborate learning mechanisms. In their model, boundedly rational strategies are not just viable but often superior, suggesting that rationality must be reinterpreted when agents face dynamic, uncertain, and interdependent settings.

The idea of using RL to represent human decision making is attracting increasing interest (Tilbury, 2022; Atashbar and Shi, 2022). Influential early contributions in this direction were the studies in Johanson et al. (2022) and Zheng et al. (2022), where deep multi-agent RL was employed to simulate emergent economic behaviors within simplified toy economies. In macroeconomics, recent literature has begun exploring RL techniques to extend classic general-equilibrium frameworks (Cook and Palmer, 2025). For instance, Mi et al. (2024) employ MARL to simulate tax policy optimization within MARL macroeconomic environments and Dwarakanath et al. (2024) employ MARL to

optimize tax credit strategies. Parallel studies focused on extending RL with macroeconomic ABMs. For instance, Brusatin et al. (2024) incorporate RL firms within a standard macroeconomic ABM, while Agrawal et al. (2025) use adversarial RL to design robust macroeconomic policies resilient to model misspecification and uncertainty. Similarly, Evans et al. (2025) propose a general framework combining adaptive agent behaviors through RL within ABMs. In Chen et al. (2021), RL is used to solve representative-agent general-equilibrium models. Similarly, in Atashbar and Shi (2023) single-agent deep RL is used to solve RBC models, and in Curry et al. (2022) RBC models are extended through MARL, while in Hill et al. (2021) RL is leveraged to solve a hybrid model that combines features of both macro ABMs and general equilibrium models. Our approach aligns with this literature by extending agent-based economic models with reinforcement learning techniques to better capture competitive interactions among firms.

3. Methods

The Poledna et al. (2023) model provides a robust agent-based framework for studying macroeconomic interactions. It focuses on how firms, households, and governments interact in various markets. One of the key strengths of the Poledna et al. (2023) model is its detailed quantitative calibration to micro, meso and macroeconomic empirical data, which makes the model capable of forecasting and policy analysis. However, to enhance its applicability in dynamic environments where firms continuously learn and adapt, we integrate a reinforcement learning mechanism (Sutton, 2018) based on independent Q-learning (IQL) into the model.

In our adaptation, each firm is modeled as an autonomous learning agent operating in a MARL environment (Albrecht et al., 2024). Unlike traditional agent-based models, including Poledna et al. (2023), where decision rules are often predefined, our approach enables firms to discover optimal pricing strategies by maximizing a reward function while interacting with the environment (Zhang et al., 2021). Firms make decisions in a dynamic, realistic environment where their pricing choices influence and are influenced by sector-wide conditions and competitors' strategies. This non-stationary environment requires firms to iteratively refine their pricing strategies based on past experiences and observed outcomes, capturing real-world adaptive behavior. Writing down the goal (in this case, maximizing the relative profit of the firm) is easier and less arbitrary than choosing the behavioral rule. In addition, the behavioral rule emerges from learning within the simulation with a data-driven mechanism that fits perfectly with the data-driven philosophy of the model of Poledna et al. (2023).

Our MARL framework follows a classic independent Q-learning (IQL) approach (Albrecht et al., 2024). Each firm learns behaviors independently without explicit coordination or direct communication with competitors. This decentralized learning structure is well-suited to all kinds of market environments. Firms optimize their pricing strategies in their environment based on past experiences rather than centralized control. Each agent maintains a Q-table, where the state-action value function is updated iteratively based on rewards received from previous pricing decisions. As a result, firms learn best-response strategies through repeated interactions with the market. The IQL enables agents to achieve a trade-off between *Exploration* (testing new pricing strategies to discover potentially better outcomes) and *Exploitation* (leveraging currently known strategies that maximize expected rewards). The reward structure is explicitly designed to reflect relative profitability, where each firm's reward is a function of its realized profit compared to the sectoral average. This formulation introduces a competitive learning dynamic, where firms continuously adjust their pricing to optimize their individual profitability and respond to sector-wide performance trends.

The description of the proposed methodology is structured around two components: the macroeconomic model (see Section 3.1) and the Q-learning design that guides firms' pricing behavior (see Section 3.2).

3.1. The model

This section outlines the core structure of the model, which builds on the framework proposed by Poledna et al. (2023). However, the model is applied to Italian data, following the calibration done by Di Domenico et al. (2025). The economy is modeled as a complex system composed of four institutional sectors: households, firms, government, and the banking sector. These sectors engage in transactions with the rest of the world through imports and exports. Firms operate within 62 sectors defined according to the NACE 2-digit classification, capturing the dynamic and interconnected nature of economic activity across industries³. Market interactions are governed by search and matching mechanisms, where the probability of a successful match depends on price competitiveness and firm size. Actual transactions are subject to budget constraints and supply availability.

Next, we present the main equations that govern agents' behavior. For a detailed description of the original model structure, refer to Poledna et al. (2023).

3.1.1. Households

Households play a dual role in the economy as consumers and suppliers of labor and capital. Consumption depends on the household's expected net disposable income, which includes social transfers. This expected income is determined by the household's labor market status and the corresponding income stream, whether from wages, expected profits, or social benefits. It also accounts for tax payments, the consumer price index from the previous period, and the expected inflation rate $\pi^e(t)$. Formally, expected income is defined as:

$$Y_h^e(t) = \begin{cases} (w_h(t)(1 - \tau^{\text{SIW}} - \tau^{\text{INC}}(1 - \tau^{\text{SIW}})) + sb^{\text{other}}(t)) \bar{P}^{\text{HH}}(t-1)(1 + \pi^e(t)) & \text{if employed} \\ (\theta^{\text{UB}} w_h(t) + sb^{\text{other}}(t)) \bar{P}^{\text{HH}}(t-1)(1 + \pi^e(t)) & \text{if unemployed} \\ (sb^{\text{inact}}(t) + sb^{\text{other}}(t)) \bar{P}^{\text{HH}}(t-1)(1 + \pi^e(t)) & \text{if not economically active} \\ \theta^{\text{DIV}}(1 - \tau^{\text{INC}})(1 - \tau^{\text{FIRM}}) \max(0, \Pi_i^e(t)) + sb^{\text{other}}(t) \bar{P}^{\text{HH}}(t-1)(1 + \pi^e(t)) & \text{if investor in firm } i \\ \theta^{\text{DIV}}(1 - \tau^{\text{INC}})(1 - \tau^{\text{FIRM}}) \max(0, \Pi_k^e(t)) + sb^{\text{other}}(t) \bar{P}^{\text{HH}}(t-1)(1 + \pi^e(t)) & \text{if bank investor} \end{cases} \quad (1)$$

where the parameter τ^{INC} represents the income tax rate, while τ^{SIW} is the employee's social insurance contribution rate, and τ^{FIRM} the corporate income tax rate. $\Pi_i^e(t)$ and $\Pi_k^e(t)$ denote the expected profits of a non-financial firm i and of the bank k , respectively, based on their previous-period performance. $\theta^{\text{DIV}} \in [0, 1]$ is the dividend payout ratio, and $\theta^{\text{UB}} \in [0, 1]$ is the unemployment-benefit replacement rate, i.e., the fraction of the worker's wage used to compute unemployment benefits. $sb^{\text{inact}}(t)$ denotes per-household social benefits for inactive households (e.g., pensions, disability or social-exclusion benefits) financed by the government, and $sb^{\text{other}}(t)$ captures additional social transfers. $\pi^e(t)$ is the expected inflation rate for period t . Finally, $\bar{P}^{\text{HH}}(t-1)$ is the consumption goods price index from the previous period.

³Products are classified according to the Classification of Products by Activity (CPA), fully aligned with NACE, with a breakdown of 64 activities. However, entries for "Services provided by extraterritorial organizations and bodies" (CPA U) contain only zeros and are excluded. Similarly, entries for "services of households as employers, undifferentiated goods and services produced by households for own use" (CPA T) are insignificant (less than 1% of the total labor factor) and are excluded.

Household consumption is modeled as a fixed share ψ of expected disposable income, net of value-added tax:

$$C_h^d(t) = \frac{\psi Y_h^e(t)}{1 + \tau^{\text{VAT}}}, \quad (2)$$

where τ^{VAT} denotes the value-added tax rate.

The resulting consumption budget is distributed across goods according to household preferences:

$$C_{h,g}^d(t) = b_g^{\text{HH}} C_h^d(t), \quad (3)$$

where b_g^{HH} represents the share of household consumption allocated to good g , as derived from the Italian Input-Output tables (Eurostat).

3.1.2. Firms

Firms operate within their respective sectors, producing goods and services by combining labor, intermediate inputs, and capital. Expectations of future demand guide their production and investment decisions.

Expected demand $Q_i^s(t)$ is formed adaptively, based on previous sales $Q_i^d(t-1)$ and anticipated macroeconomic growth $\gamma^e(t)$:

$$Q_i^s(t) = Q_i^d(t-1)(1 + \gamma^e(t)). \quad (4)$$

Production relies on a Leontief technology, implying fixed input proportions:

$$Y_i(t) = \min \{Q_i^s(t), \alpha_i N_i(t), \beta_i M_i(t-1), \kappa_i K_i(t-1)\}, \quad (5)$$

where α_i , β_i , and κ_i denote the productivity parameters for labor $N_i(t)$, intermediate goods $M_i(t-1)$, and capital $K_i(t-1)$, respectively.

Firms invest to sustain and expand their production capacity. Desired investment $I_i^d(t)$ is proportional to the depreciated portion of utilized capital:

$$I_i^d(t) = \frac{\delta_i}{\kappa_i} \min \{Q_i^s(t), \kappa_i K_i(t-1)\}, \quad (6)$$

where δ_i is the depreciation rate. Capital accumulation evolves according to:

$$K_i(t) = K_i(t-1) - \frac{\delta_i}{\kappa_i} Y_i(t) + I_i(t). \quad (7)$$

3.1.3. Government

The government sector collects taxes and redistributes income through public consumption and social transfers. Although modeled as exogenous, government behavior responds to prevailing macroeconomic conditions.

The real growth rate of government consumption $\Delta_{\%} C_G(t)$ follows an autoregressive process with a deterministic trend and stochastic component:

$$\Delta_{\%} C_G(t) = \alpha_G \Delta_{\%} C_G(t-1) + \beta_G + \epsilon_G(t), \quad (8)$$

where α_G captures the persistence of consumption growth, β_G reflects the long-term trend, and $\epsilon_G(t)$ denotes a random shock.

The government's budget balance $\Pi_G(t)$ is defined as:

$$\Pi_G(t) = Y_G(t) - (C_G(t) + B_G(t) + r_G L_G(t-1)), \quad (9)$$

where $Y_G(t)$ represents tax revenues, $C_G(t)$ is government consumption, $B_G(t)$ is total social transfers, and $r_G L_G(t-1)$ accounts for interest payments on outstanding public debt.

Public debt evolves according to the accumulation of past deficits:

$$L_G(t) = L_G(t-1) - \Pi_G(t). \quad (10)$$

3.1.4. Rest of the world

The interaction between the domestic economy and the rest of the world is captured through trade flows, specifically imports and exports. Both real imports $Y^I(t)$ and exports $C^E(t)$ are modeled using autoregressive processes applied to their respective growth rates:

$$\Delta_{\%} Y^I(t) = \alpha^I \Delta_{\%} Y^I(t-1) + \beta^I + \epsilon^I(t), \quad (11)$$

$$\Delta_{\%} C^E(t) = \alpha^E \Delta_{\%} C^E(t-1) + \beta^E + \epsilon^E(t), \quad (12)$$

where α^I and α^E capture the persistence in the growth of imports and exports, β^I and β^E reflect long-term trends, and $\epsilon^I(t)$, $\epsilon^E(t)$ are stochastic disturbances representing external shocks.

3.1.5. Banking sector and central bank

The banking sector is represented by a single commercial bank, standing in for the aggregate commercial banking system. It provides credit to both firms and households. Takes deposits from firms and households, and receives advances from (or deposits reserves at) the central bank.

At the same time, monetary policy is conducted by a central authority (the European Central Bank) that sets the policy interest rate to stabilize inflation and economic growth. The policy rate $\bar{r}(t)$ is determined according to a Taylor-type rule:

$$\bar{r}(t) = \max\left(0, \rho \bar{r}(t-1) + (1-\rho)\left(r^* + \pi^* + \xi^{\pi}\left(\pi^{\text{EA}}(t) - \pi^*\right) + \xi^{\gamma}\gamma^{\text{EA}}(t)\right)\right), \quad (13)$$

where ρ captures the degree of interest rate smoothing, r^* is the target real interest rate, and π^* denotes the target inflation. The coefficients ξ^{π} and ξ^{γ} reflect the central bank's responsiveness to inflation and output growth deviations from their targets. Inflation $\pi^{\text{EA}}(t)$ and output growth $\gamma^{\text{EA}}(t)$ in the monetary union are assumed to evolve according to autoregressive processes of order one (AR(1)).

3.1.6. Search and matching

Given the heavy computational burden imposed by repeated simulations across multiple hyperparameter configurations and initializations, the search and matching mechanism, formulated as a probabilistic process in the model of Poledna et al. (2023), has been made deterministic to speed up training significantly.

In the original probabilistic formulation, the probability of a firm being chosen depended on its price and size, as defined by:

$$\pi_i^{\text{price}} = \frac{e^{2P_i}}{\sum_{j=1}^N e^{-2P_j}} \quad (14)$$

$$\pi_i^{\text{size}} = \frac{S_i}{\sum_{j=1}^N S_j} \quad (15)$$

$$\pi_i^{\text{comb}} = \frac{\pi_i^{\text{price}} + \pi_i^{\text{size}}}{\sum_{j=1}^N (\pi_j^{\text{price}} + \pi_j^{\text{size}})} \quad (16)$$

Here, P_i denotes the price set by firm i , and S_i represents its size, measured by its total supply. The term π_i^{price} captures the relative attractiveness of firm i based on price, assigning higher values to lower-priced firms. Similarly,

π_i^{size} reflects the relative importance of firm size. The combined probability π_i^{comb} merges these two components. The final term is normalized across all firms to ensure that $\sum_i \pi_i^{\text{comb}} = 1$.

In the modified version, while the ranking still depends on these functions, it is no longer probabilistic. Instead, the ranking is now deterministic and firms sell their goods sequentially: the highest-ranked firm sells first, followed by the second-ranked, and so on. We verified that replacing the probabilistic ranking with the deterministic one does not affect the simulation outcomes. Moreover, in our implementation, the matching probability has been modified by introducing a weighted parameter α that governs the relative importance of firm size compared to price in determining match attractiveness. The resulting combined probability is defined as:

$$\pi_i^{\text{comb}}(\alpha) = \frac{\pi_i^{\text{price}} + \alpha \pi_i^{\text{size}}}{\sum_{j=1}^N (\pi_j^{\text{price}} + \alpha \pi_j^{\text{size}})}, \quad \alpha \geq 0. \quad (17)$$

This formulation allows for a flexible adjustment of the matching mechanism: when $\alpha = 0$, only price matters in the selection process; as α increases, firm size becomes more influential. The parameter thus potentially enables us to explore different market configurations, ranging from price-driven to size-driven matching dynamics. In practice, we simulate two cases: $\alpha=0$ (price-driven matching) and $\alpha=1$ (baseline case with price and size equally balanced).

This deterministic formulation, together with the adjustable parameter α , defines the environment in which firms learn and optimize their pricing behavior.

3.2. Q-learning design

We adopt an IQL algorithm to guide firms in optimizing their pricing decisions. Agents gradually learn best pricing strategies through repeated interactions with the environment, resulting in dynamic adjustments of a further markup component.

$$p_i(t) = \underbrace{p_i(t-1)}_{\text{Previous price}} \underbrace{(1 + \gamma^e(t))}_{\text{Expected inflation}} \underbrace{(1 + \pi_i(t))}_{\text{Cost-push inflation}} \underbrace{\mu_i(t)}_{\text{Price adjustment}} \quad (18)$$

By adopting IQL, each firm independently applies Q-learning within the context of its sector, maintaining a distinct Q-table and learning its optimal pricing strategy by evaluating performance relative to its sectoral peers. During training, Q-tables are updated cumulatively over multiple epochs, each consisting of T periods, after which the model is re-initialized to its initial state. This iterative process allows firms to accumulate experience across different realizations of the economy, progressively enhancing the robustness and effectiveness of their decision-making strategies. Firms operate within sectors that differ in the number of firms and total employment based on data classified according to NACE codes. Full details—including the number of firms and employment levels per sector—are provided in tables A.11 and A.12. In the original model, firms' employment and size are randomly assigned in each simulation according to a power-law distribution, leading to variability across runs. In our training framework, by contrast, each firm in the model retains fixed characteristics—such as its size—throughout all training epochs. Although the macroeconomic environment evolves differently across Monte Carlo simulations, firm-specific features remain stable. For example, firm 1 is consistently the largest in every run. This design allows us to identify better how a firm's optimal pricing policy depends on its structural characteristics.

3.2.1. Learning process

Each firm maintains and updates its own Q-table $Q_i(s, a)$ following the standard Q-learning update rule:

$$Q_i(s_{i,t}, a_{i,t}) \leftarrow Q_i(s_{i,t}, a_{i,t}) + \alpha_Q \left(r_{i,t} + \gamma_Q \max_{a'_{i,t}} Q_i(s_{i,t+1}, a'_{i,t}) - Q_i(s_{i,t}, a_{i,t}) \right). \quad (19)$$

The updating rule is based on the Bellman equation, which allows the agent to recursively estimate the optimal value of the state-action function Q .

- $Q_i(s_{i,t}, a_{i,t})$ is the estimated value for firm i of taking action $a_{i,t}$ in state $s_{i,t}$,
- α_Q is the learning rate, determining the speed of Q-value updates,
- $r_{i,t}$ is the immediate reward received by firm i after taking action $a_{i,t}$,
- γ_Q is the discount factor, weighting future rewards relative to immediate ones,
- $\max_{a'} Q_i(s_{i,t+1}, a'_{i,t})$ represents the maximum expected future return for firm i at the next state $s_{i,t+1}$, considering all possible actions a' .

In this equation, the term $r_{i,t} + \gamma_Q \max_{a'_{i,t}} Q_i(s_{i,t+1}, a'_{i,t})$ represents the target, which is the immediate reward plus the discounted value of the best future action in the next state. The difference $r_{i,t} + \gamma_Q \max_{a'} Q_i(s_{i,t+1}, a'_{i,t}) - Q_i(s_{i,t}, a_{i,t})$ is the Temporal Difference (TD) error, which is used to update the current estimate $Q_i(s_{i,t}, a_{i,t})$. The goal of the Q-learning algorithm is to minimize this error over time by updating the Q-values.

3.2.2. Observation space and action space

The observation space of each firm captures the relative deviation of its price from the average price in its sector. Specifically, for firm i at time t :

$$s_{i,t} = \left(\frac{p_{i,t} - \bar{p}_{s(i),t}}{\bar{p}_{s(i),t}} \right), \quad (20)$$

where $p_{i,t}$ denotes the firm's price and $\bar{p}_{s(i),t}$ is the sectoral average price. The resulting value is then mapped to a discrete state space by grouping into predefined intervals.

The action space of firms is represented by a discrete set of choices among a discrete set of multiplicative price adjustment factors (e.g., from 0.95 to 1.05). Each action $a_{i,t}$ updates the price according to the adjustment factor $\mu_i(\cdot)$ associated with the selected action.

3.2.3. Exploration policy

Agents follow an ϵ -greedy strategy to balance exploration and exploitation: with probability ϵ_Q , an agent explores by selecting a random action, while with probability $1 - \epsilon_Q$, it exploits its current knowledge by choosing the action that maximizes the Q-value. Given the multi-agent setting, we tested both a constant ϵ_Q , to preserve exploration throughout training, and a linearly decaying ϵ_Q , which progressively favors exploitation as learning advances, an approach more in line with standard reinforcement learning practices.

3.2.4. Reward function

The reward $r_{i,t}$ reflects the firm's profit rate relative to the average profit rate in its sector and is defined as:

$$r_{i,t} = \pi_{i,t} - \bar{\pi}_{s(i),t} \quad (21)$$

where $\pi_{i,t}$ denotes the firm's profit rate and $\bar{\pi}_{s(i),t}$ the sectoral mean. Following is an explanation of the method we use to calculate the profit rate.

In the original model, profit is calculated as:

$$\begin{aligned} \Pi_i(t) = & \underbrace{P_i(t)Q_i(t)}_{\text{Sales}} + \underbrace{P_i(t)\Delta S_i(t)}_{\text{Inventory change}} - \underbrace{(1 + \tau^{\text{SIF}})w_i(t)N_i(t)\bar{P}^{\text{HH}}(t)}_{\text{Labour costs}} \\ & - \underbrace{\frac{1}{\beta_i}\bar{P}_i(t)Q_i^I(t)}_{\text{Material costs}} - \underbrace{\frac{\delta_i}{\kappa_i}P_i^{\text{CF}}(t)Q_i^K(t)}_{\text{Depreciation}} \\ & - \underbrace{\tau_i^Y P_i(t)Y_i(t) - \tau_i^K P_i(t)Y_i(t)}_{\text{Net taxes/subsidies on products/production}} \\ & - \underbrace{\hat{r}(t)(L_i(t-1) + \max(0, -D_i(t-1)))}_{\text{Interest payments}} + \underbrace{\bar{r}(t)\max(0, D_i(t-1))}_{\text{Interest received}} \end{aligned} \quad (22)$$

where $\hat{r}(t)$ and $\bar{r}(t)$ are the interest rates paid on outstanding loans and received on deposits. $\bar{P}_i(t)$ and $P_i^{\text{CF}}(t)$ are the actual prices paid by firm i for intermediate goods and investment in capital goods. $\bar{P}^{\text{HH}}(t)$ is the consumer price index (CPI). τ_i^Y is the net tax rate on products, while τ_i^K is the net tax rate on production. $D_i(t)$ represents the firm's deposits, which can generate interest income when positive.

However, this calculation does not represent the monetary profit capitalists actually realize through the process of producing goods and services, which is what they seek to maximize. Instead, we apply a version of EBITDA (Earnings Before Interest, Taxes, Depreciation, and Amortization). Specifically, we introduce the following adjustments to Equation 22:

1. Removal of the inventory change term.
2. Inclusion of intermediate goods purchased during the period (not only those used).
3. Exclusion of depreciation costs.
4. Exclusion of taxes and subsidies.
5. Exclusion of interest (both paid and received).

The profit rate $\pi_{i,t}$ is then calculated by dividing the adjusted profit by the value of capital $K_i(t)$, which includes both the physical capital stock and the value of inventories.

In some simulations, we also tested a *penalized reward* function that amplifies losses relative to the sectoral mean. Specifically:

$$r_{i,t} = \begin{cases} 2 \cdot (\pi_{i,t} - \bar{\pi}_{s(i),t}) & \text{if } \pi_{i,t} < \bar{\pi}_{s(i),t} \\ \pi_{i,t} - \bar{\pi}_{s(i),t} & \text{otherwise} \end{cases} \quad (23)$$

This formulation preserves the direction of the deviation while penalizing underperformance more heavily.

3.2.5. Experimental procedure

Building on the calibration of the model developed by Di Domenico et al. (2025), we exploit multiple calibration windows to enrich the training process. Each calibration window provides a distinct initialization, allowing the model to evolve dynamically from different starting macroeconomic conditions. As a result, training epochs are executed across diverse initializations, enhancing the robustness of the learning process and mitigating the risk of overfitting to a particular trajectory.

Between 2011Q1 and 2013Q4, the training phase is carried out using a 12-quarter rolling window scheme, with each initialization based on real Italian data. We adopt a dual-track strategy. The first track (see Section 4.1), implemented in a single-to-multi-agent setting (using a model scaled from 1 agent to 100,000 agents), validates the algorithm’s ability to learn and outperform the baseline decision rule. The second track (see Section 4.2), conducted in a complete multi-agent setting (with a model scaled from 1 agent to 1,000 agents), aims to extract optimal pricing strategies and assess their stability across agents. In this second configuration, all firms learn simultaneously, enabling us to evaluate collective dynamics and prepare the model for forecasting applications.

The out-of-sample validation phase exploits the 2014Q1 initialization, with simulations running for 12 quarters, i.e., up to 2016Q4. In the single-to-multi-agent track, validation relies on improvements in firm-level profits relative to a baseline pricing behavior. In the multi-agent track, validation focuses on the ability of the learned strategies to replicate observed inflation dynamics. This validation is complemented by the computation of an *action stability index*, which captures the consistency of the learned policies over time. This phase thus allowed us to select the model configuration, in terms of hyperparameters, that best reproduces the empirical inflation dynamics observed in the Italian economy.

To assess predictive performance on a previously unused time span, the forecasting phase covers the period from 2017Q1 to 2019Q4. The best-performing models, identified through the validation exercises, are deployed for out-of-sample forecasting on five macroeconomic variables. The forecasts are then compared with those developed from simulations without firms capable of using RL and with respect to the basic econometric model represented by AR(1), allowing us to evaluate the model’s predictive accuracy.

4. Results: mean reversion of prices

In Section 4 and 5, we present the main findings of our study. The analysis proceeds in four steps. First, in Section 4.1, we examine the transition from single- to multi-agent learning, which provides the basis for validating the proposed algorithm, assessing the algorithm’s effectiveness in enabling firms to achieve above-sector-average profits. Second, in Section 4.2, we investigate the dynamics of multi-agent optimization learning, highlighting the emergence of stable strategies and adaptive mean-reverting behaviors. Third, in Section 5 we evaluate the forecasting performances of the proposed reinforcement learning framework in comparison with the benchmark approach. Finally, in Section 5.1, we complement these results with standard OLS regression estimates, offering a conventional statistical benchmark against which the predictive performance of the learning algorithms can be assessed.

4.1. Single to multi agent learning - Algorithm validation

In our multi-agent learning framework, agents observe price differences relative to their competitors and determine their “optimal” actions according to the reward function. To validate the learning process, we compare the outcomes

after training with a baseline scenario where no machine learning is applied. This assessment determines whether the trained agents effectively maximized their performance, as they discover their “optimal” actions. Specifically, we analyze the profits of trained agents against those of the same agents when they are naive. Due to the substantial computational effort required, we initiate our analysis using a small-scale model, scaled from 1 to 100,000. Given that learning is a comparative process, we focus on the most populated sector, which is the 30th sector, namely *CPA_G47* (Retail trade services, except for motor vehicles and motorcycles). According to the selected scale, this sector comprises six firms, which, after rounding, have unique identifiers ranging from 40 to 45 (for an illustration of the distribution of firms and employees across 62 sectors for the different scales, see Table A.11).

Initially, we train agents individually in controlled, single-agent environments to establish performance baselines, capturing their isolated behavior and profit-generating capabilities under consistent training parameters. Subsequently, we train agents within progressively larger multi-agent environments (i.e., with increasing group sizes) to evaluate their interactions and the dynamics that emerge when agents optimize simultaneously. This phase allows us to observe how agents influence each other’s performance. To comprehensively evaluate the learning process and avoid bias due to the specificity of individual agents, we conduct simulations for every possible combination of these six agents across different group sizes. This approach allows us to observe the transition from single-agent to multi-agent environments, enabling a thorough comparison of optimized actions and profit outcomes against a predefined baseline.

In total, we examine 63 unique combinations, calculated as the sum of combinations for each learning group size:

$$\sum_{k=1}^6 \binom{6}{k} = 6 + 15 + 20 + 15 + 6 + 1 = 63$$

These combinations are detailed in Table A.10, where each row corresponds to a different learning group size and lists the specific agent IDs involved in each combination. For each combination, we conduct a series of training (2011Q1-2013Q4) and evaluation (2014Q1-2016Q4) phases to assess the performance of our learning process, as explained in Section 3.2.5.

In Figure 1, the solid black lines represent the deviation from the average sectoral profit rate achieved on average by an RL-performing agent across all possible group combinations (order being irrelevant). In contrast, the dotted black lines represent the corresponding average performance of their competitors. Shaded areas indicate 90% confidence intervals (of the averages), illustrating variability and statistical significance.

The graph shows that agents using reinforcement learning outperform those following baseline behavior. When all agents apply RL, the group comprises the full population, and the average deviation from the mean is, by construction, zero. As the number of RL agents increases, the relative underperformance of baseline agents becomes more pronounced. These results confirm that the RL algorithm operates correctly, identifying and exploiting profit-maximizing strategies relative to baseline behavior.

To clearly understand the experimental setup, Table 1 summarizes the specific hyperparameters used in the single-to multi-agent learning model.

Figure 1: Mean profits of optimizing agents vs baseline across different group sizes

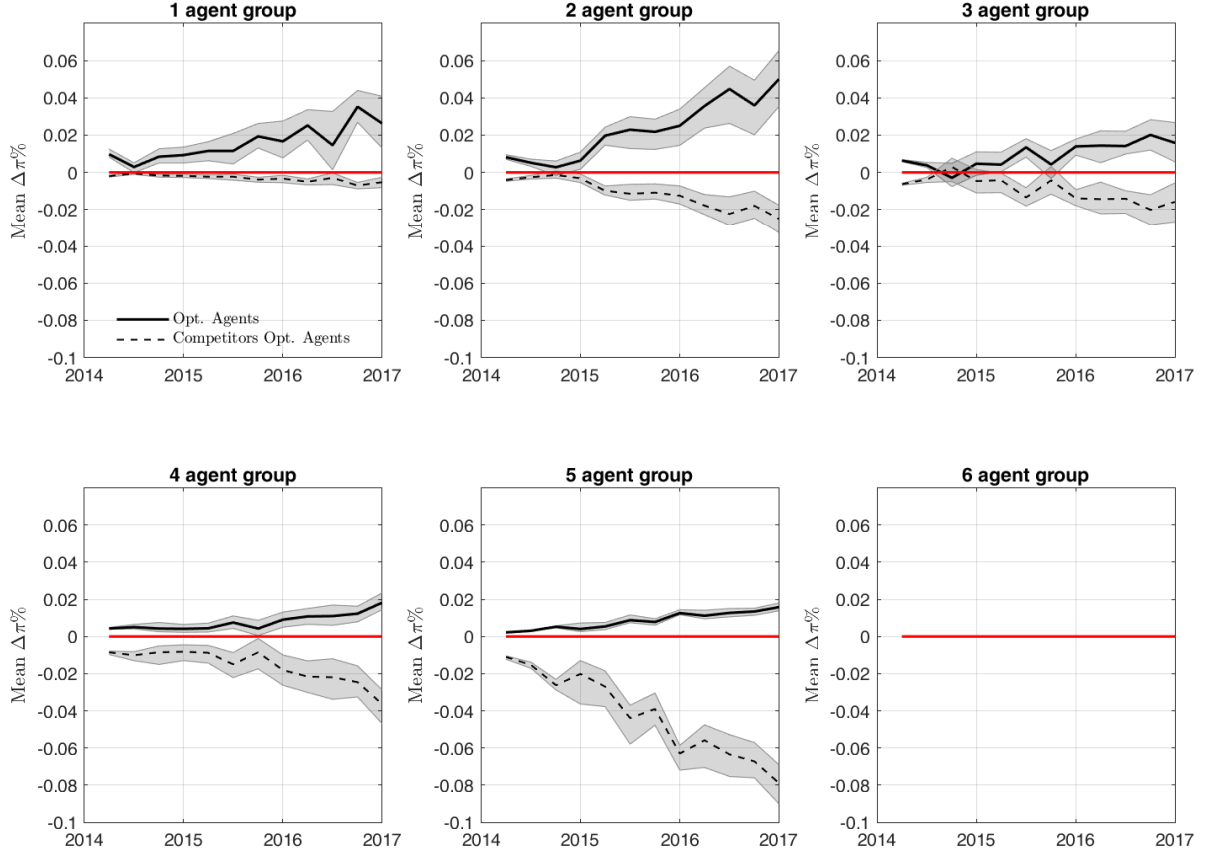


Table 1: Parameter settings for the single to multi agent learning model

Hyperparameter	Values	Description
n_{states}	5	Number of discrete states defining the observation space.
$n_{actions}$	5	Number of discrete actions available to firms when adjusting their price.
Δp	$[-0.1, 0.1]$	Price change bounds (state discretization limits). Combined with $n_{states} = 5$, this results in a discretization step of 0.04, generating 5 bins: $[-0.10, -0.06]$, $[-0.06, -0.02]$, $[-0.02, 0.02]$, $[0.02, 0.06]$, $[0.06, 0.10]$.
ϕ_{min}, ϕ_{max}	$[0.90, 1.10]$	Minimum and maximum multipliers used for price adjustments. Combined with $n_{actions} = 5$, this defines a discretized action space with a step size of approximately 0.05, yielding the following multiplicative factors: 0.90, 0.95, 1.00, 1.05, 1.10.
$\epsilon_{start}, \epsilon_{end}$	$[0.95, 0.1]$	Initial and final exploration rates in the ϵ -greedy policy, decaying over training epochs.
Learning rate (α_Q)	0.1	Step size used to update Q-values.
Discount factor (γ_Q)	0.99	Relative importance of future versus immediate rewards in Q-learning.
E_{train}	300	Number of training epochs.
E_{test}	500	Number of testing epochs.
T	12	Periods per epoch.

4.2. Multiagent optimization learning

Between 2011Q1 and 2013Q4 all firms are jointly trained. Optimal pricing strategies are derived, and the stability of their behavior is assessed through a *stability index*. The model is then validated using 2014Q1 data, with the best-performing specification selection based on its predictive accuracy for inflation over the targeted time horizon.

4.2.1. Stabilization

We computed a Stability Index (SI) for each agent and state over time to analyze the stability of agents' optimal actions during the learning process. This approach allows us to observe whether agents converge to consistent strategies. The stability index measures how frequently the optimal action for a specific state changes over time, acting as a proxy for policy convergence (Albrecht et al., 2024). Fewer changes indicate higher stability, while more frequent changes suggest lower stability. The stability index is computed by counting the times the optimal action at time t differs from that at time $t - 1$. The formula is given by:

$$SI_t(s) = 1 - \frac{\sum_{t'=t}^{t+T} \mathbb{1}_{\{a_{t'}(s) \neq a_{t'-1}(s \pm 1)\}}}{T} \quad (24)$$

where $a_t(s)$ is the optimal action at time t for state s , T is the selected time window, and $\mathbb{1}_{\{\cdot\}}$ is an indicator function that equals one if the actions differ by more than one step, and zero otherwise. The SI is a value between 0 and 1, and 1 indicates perfect stability (no changes in actions), while a value closer to 0 indicates frequent changes in actions, signifying low stability. States near the center of the state space (in particular, states 5, 4, and 6) tend to be less stable than peripheral states, although their SI values remain high. To illustrate this concept, Figure 2 shows the evolution of the SI across the training phase based on a selected simulation scenario. In this case, firm size affects the search and matching process ($\alpha = 1$), and price deviations range between -2% and $+2\%$, while the exploration rate ϵ decays linearly down to 0.5. The training environment has 8 states and 8 actions. Each line shows the stability of the chosen action when the agent is in a given state.

Figure 2 shows that the choice of the agents is very stable during the training phase. Similar results are found in the other parameter settings.

4.2.2. Testing and model validation

Validating trained models is necessary to ensure that learning processes do not lead to overfitting or economically implausible behaviours. Especially in multi-agent settings, validation helps assess whether the strategies emerging from interaction and adaptation are consistent, robust, and capable of reproducing key macroeconomic patterns. In our case, we first evaluated the stability and economic plausibility of agents' optimal actions and robustness across simulations. This step allowed us to identify a subset of well-performing models among a broader set trained under varying hyperparameters. Table 2 includes the ranges of the hyperparameters tested, both in terms of bounds and intervals (e.g. different action and state ranges, learning rates and exploration rate).

Figure 2: Evolution of the Stability Index across the training

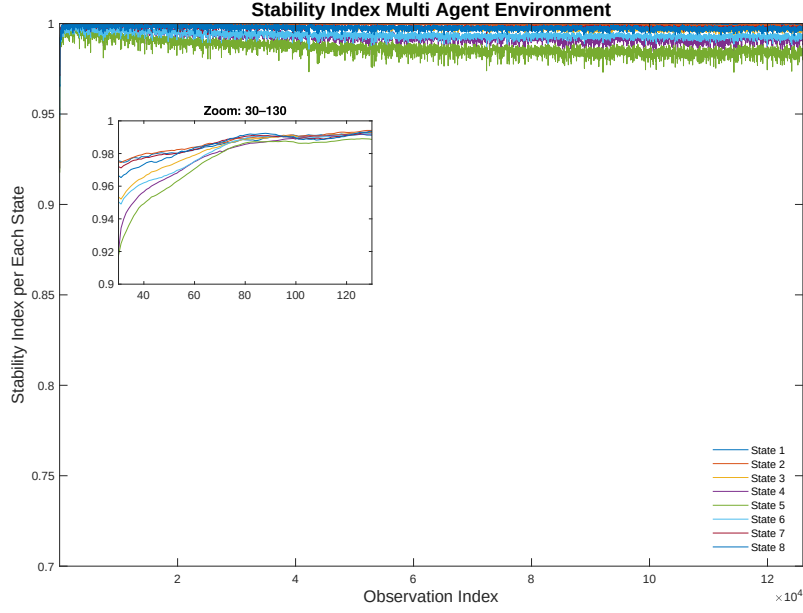


Table 2: Hyperparameter space used during model training

Hyperparameter	Values tested	Description
n_{states}	Discrete values in [4, 20]	Number of discrete states defining the observation space. All values were tested with unit step to explore the effect of state granularity on learning.
n_{action}	Discrete values in [4, 20]	Number of discrete actions available to firms when adjusting their price. All values were tested with unit step to explore the effect of action granularity.
Δp	Grid within $[-0.1, 0.1]$	Minimum and maximum price change bounds (state discretization limits). All symmetric ranges with step 1% (from $\pm 1\%$ to $\pm 10\%$) were tested to define firms' state space.
$\phi_{\min}, \phi_{\max}$	Grid within [0.90, 1.10]	Minimum and maximum price adjustment factors. Various symmetric ranges (e.g., $\pm 1\%$, $\pm 2\%$, $\pm 3\%$, ..., $\pm 10\%$) were tested to define firms' action space.
$\epsilon_{\text{start}}, \epsilon_{\text{end}}$	Multiple pairs in [0.95, 0.1]	Exploration rate schedule for the ϵ -greedy policy. Several combinations of initial and final values were tested to control the speed and extent of convergence.
α	{0, 1}	Weight of firm size in the search and matching process. When $\alpha = 1$, larger firms are more likely to be matched.
M	{my, myp}	Reward function: my applies a linear reward function while myp applies a non-linear penalization for negative profit rates deviations (see Equation 23).
α_Q	Grid within [0.01, 0.1]	Step size used to update Q-values. All values were tested with step size 0.01 to balance learning speed and convergence stability.
γ_Q	Values in [0.90, 0.99]	Determines the relative importance of future versus immediate rewards in Q-learning. All values were tested with step size 0.01 to balance foresight and reactivity.

We then assess the ability of this subset to replicate observed inflation dynamics over the 2014Q1–2016Q4 validation period across 12 quarters.

We decided to retain the four best-performing models regarding inflation forecasting accuracy. Their performance is reported in Table 3, which reports the percentage change in Root Mean Squared Error (RMSE) relative to the AR(1)

benchmark model. A positive (negative) sign indicates an improvement (worsening) in forecasting performance.

Among the four selected models, *Mod(2)* shows the best performance in terms of inflation forecasting. Its key features include a narrower range of pricing actions, with $\phi_{\min} = 0.98$ and $\phi_{\max} = 1.02$, meaning that firms can only adjust their prices within $\pm 2\%$; a decaying exploration rate, with ϵ decreasing from 0.9 to 0.5; and a search and matching process that also depends on firm size, captured by $\alpha = 1$.

Mod(4), the second-best model, shares the same price adjustment range and exploration decay as *Mod(2)*, but removes the influence of firm size from the search and matching mechanism by setting $\alpha = 0$. This simplification slightly weakens the model’s ability to coordinate firm behavior in a way that replicates observed inflation dynamics.

Mod(3) ranks third. It allows for a wider range of price adjustments, with $\phi_{\min} = 0.97$ and $\phi_{\max} = 1.03$, and keeps a constant exploration rate at $\epsilon = 0.9$, while maintaining $\alpha = 1$.

Finally, *Mod(1)* is the least accurate among the four. It shares *Mod(3)*’s price range and constant exploration rate but sets $\alpha = 0$, thus preventing firm size from influencing the search and matching process.

Table 3: Out-of-sample forecast performance (RMSE) of inflation compared to the AR(1) model. The testing period is 2014Q2 to 2016Q4. The AR(1) model is estimated each quarter starting in 2014Q2 using time series data from 1996Q1. ABM results are based on the average of 200 Monte Carlo simulations initialized quarterly between 2014Q1 and 2016Q4.

	<i>Inflation</i>			
	Mod(1)	Mod(2)	Mod(3)	Mod(4)
1q	-297.3	65.5	-211.6	-119.3
2q	-324.4	97.3	-190.3	81.3
4q	-380.7	91.9	-201.0	89.4
8q	-203.4	25.8	-22.7	5.6

$Mod1 = \phi_{-0.03_0.03_nstates_8_e_{0.9_0.9_}\alpha_{0.1_}\gamma_{0.99_M_my_}\alpha1}$
 $Mod2 = \phi_{-0.02_0.02_nstates_8_e_{0.9_0.5_}\alpha_{0.1_}\gamma_{0.99_M_my_}\alpha1}$
 $Mod3 = \phi_{-0.03_0.03_nstates_8_e_{0.9_0.9_}\alpha_{0.1_}\gamma_{0.99_M_myp_}\alpha1}$
 $Mod4 = \phi_{-0.02_0.02_nstates_8_e_{0.9_0.5_}\alpha_{0.1_}\gamma_{0.99_M_myp_}\alpha0}$
 ABM percentage improvements (+) or losses (−) relative to AR(1).

4.2.3. Selected actions

In this section, we report the analysis of the actions selected by agents trained with the MARL mechanisms previously explained. In Figures 3 and 5 size does not influence the search and matching process ($\alpha = 0$), while in Figures 4 and 6 size matters. Price deviations range between -3% and $+3\%$ in simulations reported in figures 3 and 4 where the exploration rate ϵ is kept constant throughout the simulation. Instead, price deviations range between -2% and $+2\%$ in simulations reported in Figures 5 and 6 where the exploration rate ϵ decays linearly down to 0.5 throughout the simulation. Firms observe their state, which is reported in the x-axis, where 1 indicates the highest negative price deviation and 8 indicates the highest positive price deviation from the sectoral average. The selected action is reported on the y-axis, where 1 corresponds to the highest price decrease and 8 corresponds to the highest price increase. Each column includes sectors with a certain number of firms (S), starting from the first column on the left, reporting sectors having up to 5 firms, to the last column on the right, reporting sectors with more than 300 firms. The rows divide firms based on their market share (F), starting from the top row with firms holding a market share below 0.5%, down to the bottom row with firms holding a market share above 45%. The color of the dot becomes darker as the number of firms choosing that action increases relative to the total number of firms in the economy. The size of the dot increases with the number of firms choosing that action relative to the total number of firms in that

subplot.

As already said, figures 3 and 5 report the results of simulations in which size does not influence the search and matching process. A clear tendency toward mean-reverting behavior can be observed: when a firm is in a low-price state (low values on the x-axis), it tends to choose actions that increase the price (high values on the y-axis), and vice versa when the firm is in a high-price state (high values on the x-axis), it tends to choose actions that lower the price (low values on the y-axis), forming a sort of diagonal pattern across the various subplots. In Figure 3, when ϵ is kept constant, the diagonal appears higher in the subplots, indicating a general tendency toward price increases. In contrast, in Figure 5, when ϵ decays, the diagonal shifts lower, indicating a general tendency toward price reductions.

Figures 4 and 6 report the results of simulations in which size influences the search and matching process. When ϵ is kept constant (Figure 4), the mean-reverting tendency is confirmed, as evidenced by the formation of the diagonal, clearly visible in the top-right subplots, where the majority of firms are concentrated. When ϵ decays (Figure 6), instead we observe that smaller firms tend to lower their prices to maintain or increase their market share, while larger firms, which can benefit from their size, exhibit a much less defined behavior.

In general, the lower-right quadrant of the figures remains sparsely populated, reflecting the empirical regularity that sectors with many firms rarely feature dominant players. When firm size influences the search and matching process, the relationship between previous relative price and price adjustment becomes more nuanced. Smaller firms—particularly in highly populated sectors—tend to lower their prices more aggressively to defend or expand their market share, shifting the mean-reverting diagonal slightly downward. In contrast, firms with large market shares operating in less crowded sectors, visible in the bottom-left areas, display more dispersed behaviors. Here, the diagonal pattern fades, suggesting that some firms may leverage their market power and deviate from the mean-reverting rule. Enabling size-based matching introduces heterogeneous behavioral regimes: competitive pressure is stronger in fragmented sectors, while dominant firms in concentrated markets enjoy greater strategic flexibility.

Interestingly, the pricing policy that our RL firms tend to learn, letting prices revert to the market average, has long been presented as a practical heuristic in macro-ABMs. Early implementations, most notably the original family of so-called ‘CATS’ models (Delli Gatti et al., 2011; Assenza et al., 2015), already used the cross-sectional average price as a pivot: when a firm’s price sits above the market average and inventories are high, it cuts price; when it is below the average and shortages emerge, it raises price.

Later extensions have preserved and refined this heuristic adjustment, such as in the models of Wiese et al. (2024), Hommes et al. (2025), and in the so-called ‘Mark-0’ model (Gualdi et al., 2015).

In this sense, our result echoes a robust behavioral rule that has been repeatedly embedded — sometimes implicitly, sometimes by construction — across several macro-ABM frameworks.

Moreover, empirical studies in industrial economics and applied microeconomics corroborate the prevalence of such reference-based pricing. Large-scale experiments show that firms adjust their prices towards competitors’ averages when provided with market information (Kim, 2024). Survey evidence confirms that a substantial share of firms explicitly declare pricing with reference to competitors (Fabiani et al., 2006), and that most European manufacturers follow full-cost pricing with sector-standard profit margins (Altomonte et al., 2015). At the firm level, profitability displays strong mean reversion towards sectoral benchmarks (Fama and French, 2000). At the same time, evidence from fragmented retail markets reveals that local firms often match the prices of selected reference competitors, generating domino effects in price dynamics (Atkinson et al., 2009). Together, these findings suggest that *reference pricing* and *profit mean-reversion* are both empirically widespread and consistent with the heuristics adopted in macro-ABMs,

and with the strategies endogenously learned by our RL firms.

In conclusion, our study contributes a novel theory of price-setting in complex markets, one that is not assumed a priori but emerges from the interplay of heterogeneity and learning. The mean-reverting pricing rule we identify can be seen as an endogenous competitive strategy that blends elements of classical theory (firms respond to excess prices by cutting them and vice versa) with real-world behavioral elements (firms use heuristics based on peer prices). This strategy stands in contrast to DSGE models (Clarida et al., 1999; Galí and Gertler, 1999), where optimal pricing is typically derived from a first-order condition, and to traditional oligopoly theory, where equilibrium assumes common knowledge of strategies. Instead, our approach shows how quasi-rational agents can dynamically “discover” a pricing rule that yields coordination (prices gravitating toward an industry average) without any central direction. By doing so, we provide a price formation model that resonates with empirical observations (firms often follow competitors’ pricing leads) and is grounded in adaptive optimization rather than pure assumption.

Figure 3: Firm size does not influence the search and matching process ($\alpha = 0$). Price deviations range between -3% and $+3\%$. The exploration rate ϵ is kept constant throughout the simulation. Y-axis: Optimal actions, where 8 corresponds to the highest price increase. X-axis: Firm observed states, where 8 indicates the highest positive price deviation from the sectoral average. Rows: Increasing firm market share (F). Columns: Increasing number of firms in the sector (S). Dot color: Relative number of firms in the entire economy. Dot size: Relative number of firms within each subplot.

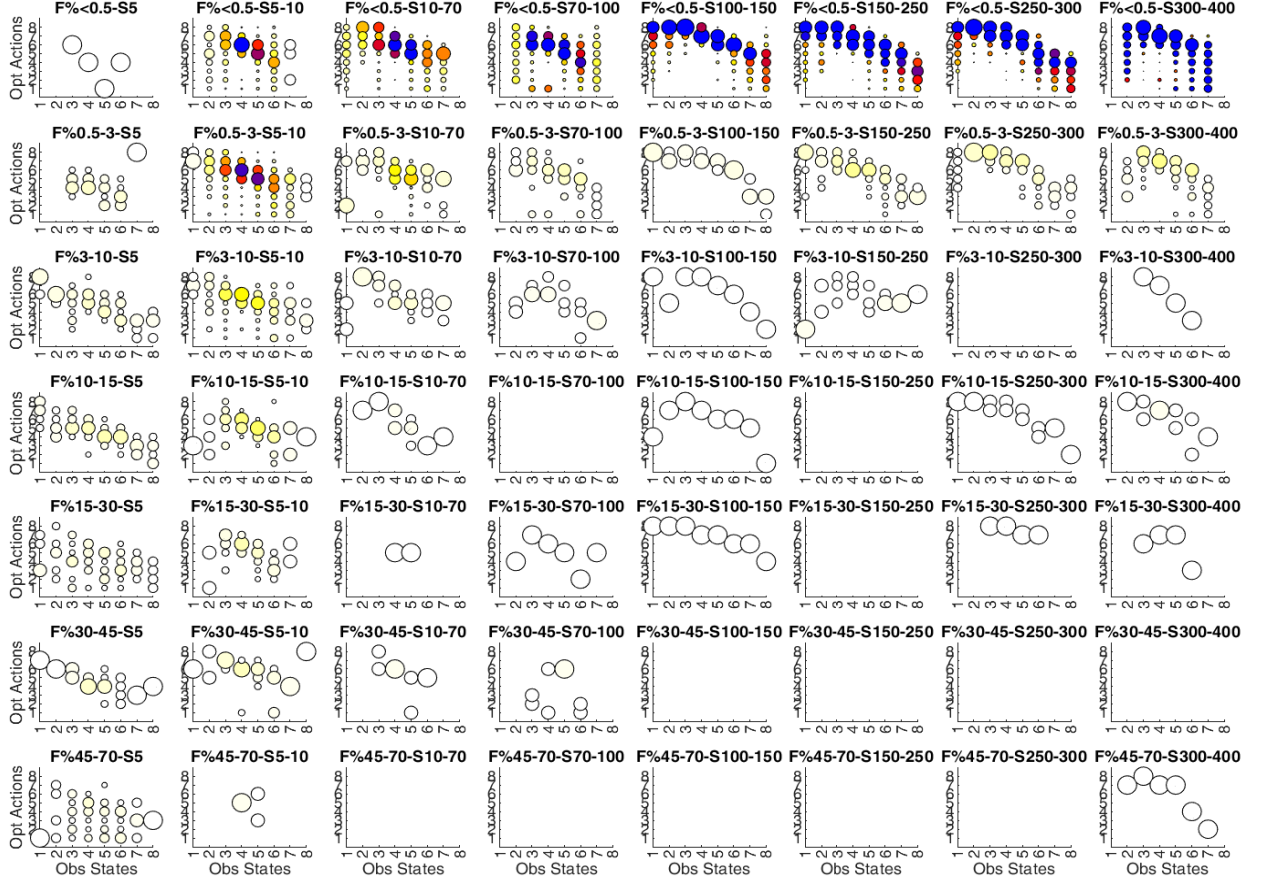


Figure 4: Firm size influences the search and matching process ($\alpha = 1$). Price deviations range between -3% and $+3\%$. The exploration rate ϵ is kept constant throughout the simulation. Y-axis: Optimal actions, where 8 corresponds to the highest price increase. X-axis: Firm observed states, where 8 indicates the highest positive price deviation from the sectoral average. Rows: Increasing firm market share (F). Columns: Increasing number of firms in the sector (S). Dot color: Relative number of firms in the entire economy. Dot size: Relative number of firms within each subplot.

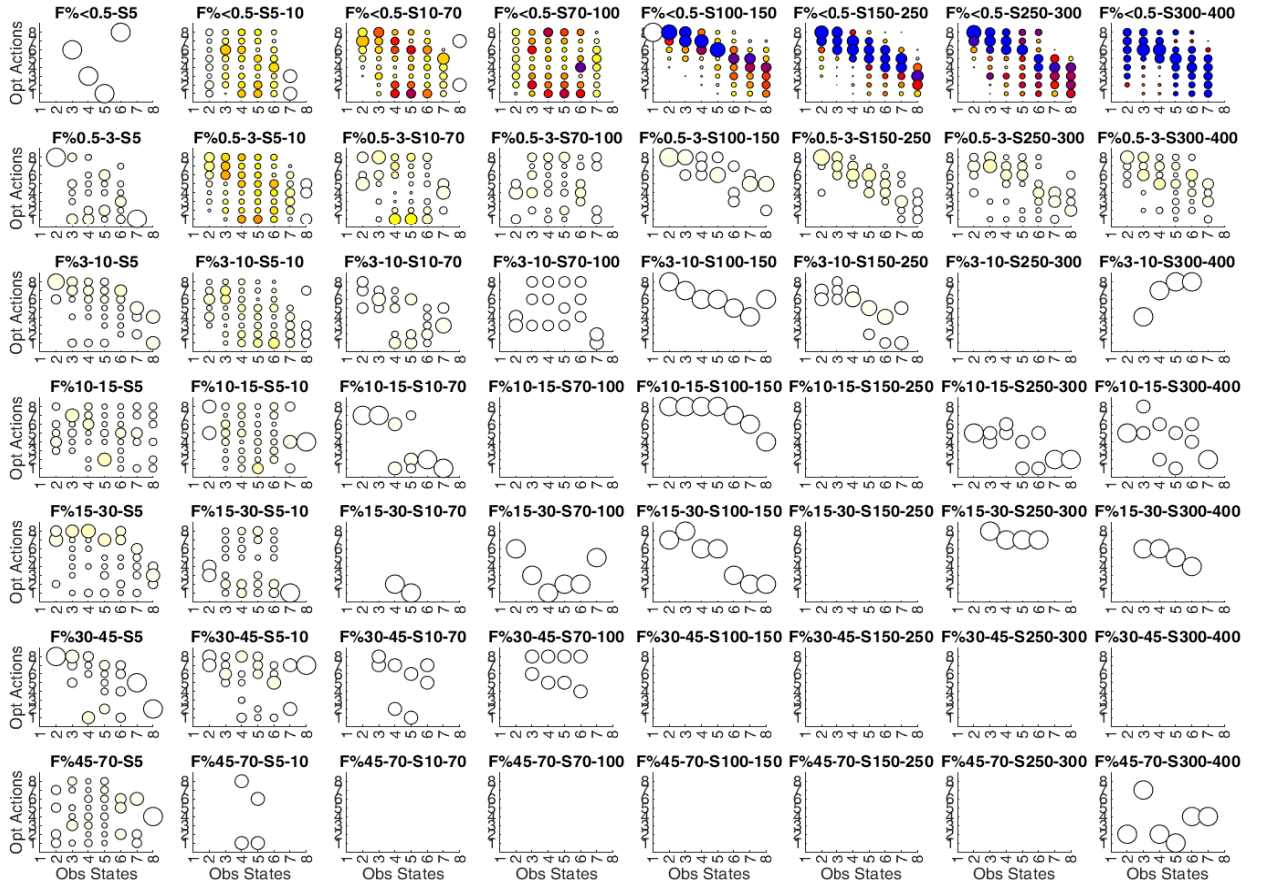


Figure 5: Firm size does not influence the search and matching process ($\alpha = 0$). Price deviations range between -2% and $+2\%$. The exploration rate ϵ decays linearly down to 0.5. Y-axis: Optimal actions, where 8 corresponds to the highest price increase. X-axis: Firm observed states, where 8 indicates the highest positive price deviation from the sectoral average. Rows: Increasing firm market share (F). Columns: Increasing number of firms in the sector (S). Dot color: Relative number of firms in the entire economy. Dot size: Relative number of firms within each subplot.

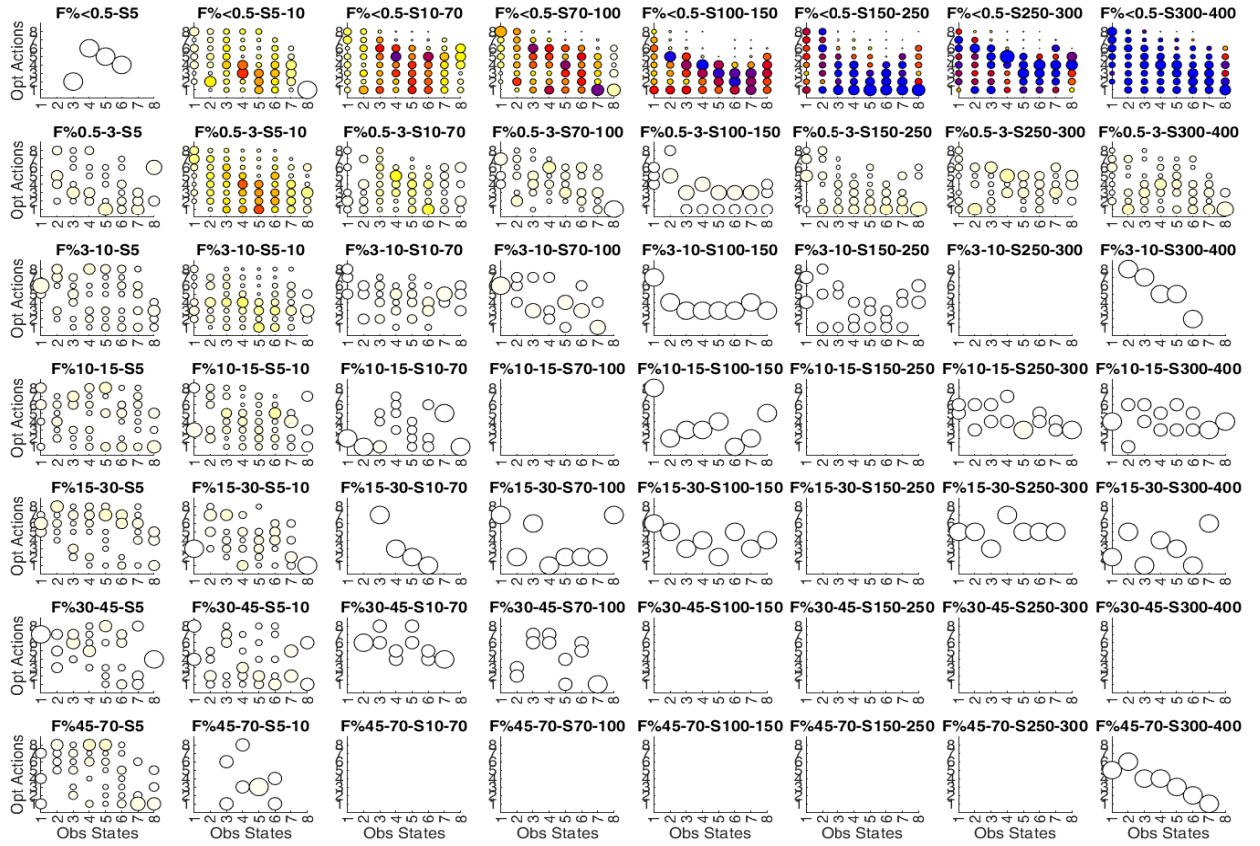
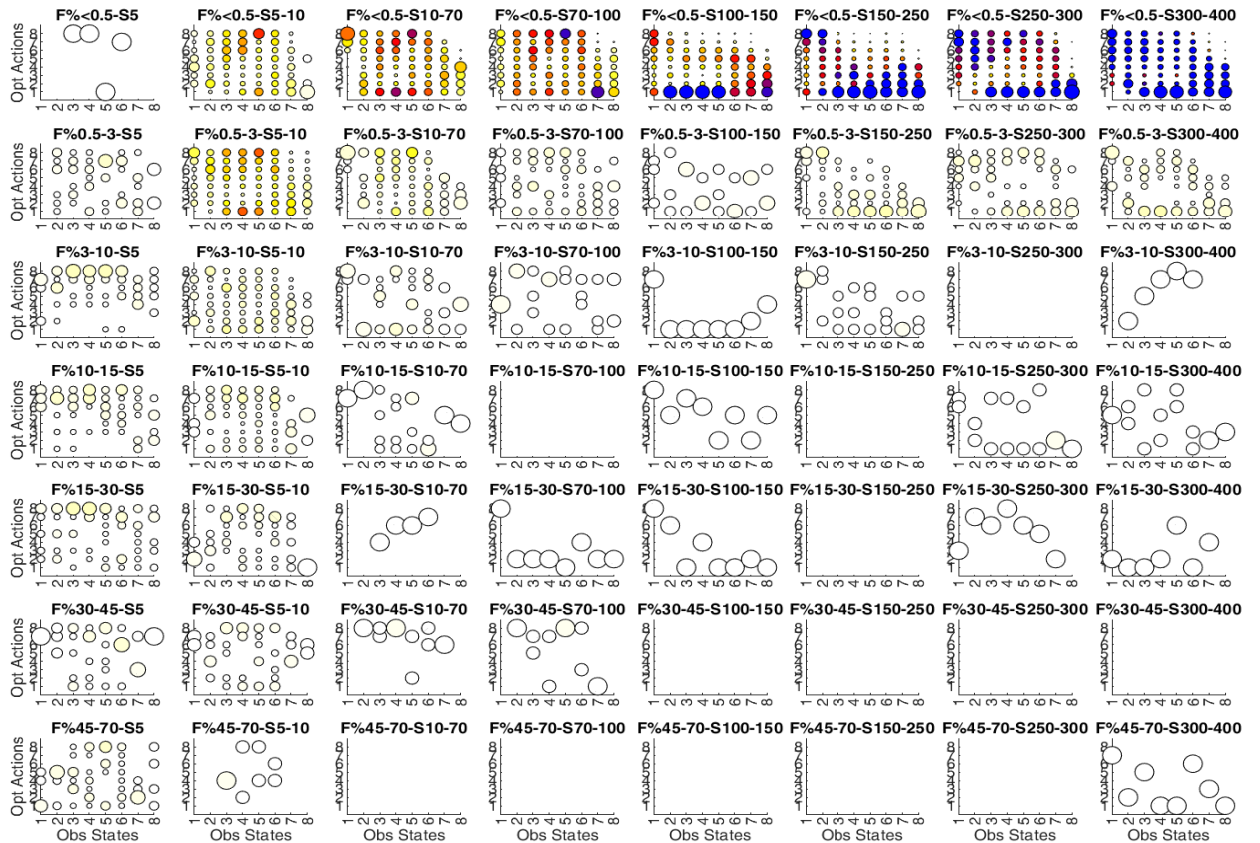


Figure 6: Firm size influences the search and matching process ($\alpha = 1$). Price deviations range between -2% and $+2\%$. The exploration rate ϵ decays linearly down to 0.5. Y-axis: Optimal actions, where 8 corresponds to the highest price increase. X-axis: Firm observed states, where 8 indicates the highest positive price deviation from the sectoral average. Rows: Increasing firm market share (F). Columns: Increasing number of firms in the sector (S). Dot color: Relative number of firms in the entire economy. Dot size: Relative number of firms within each subplot.



5. Results: forecasting performance

We take the four models selected in the previous section and run simulations with the following shares of firms implementing the optimal actions derived from reinforcement learning: 0%, 25%, 50%, 75%, and 100%. The simulation period spans 2017Q2 to 2019Q4, while the initialization window covers 2017Q1 to 2019Q3. The forecasted macroeconomic variables include GDP, inflation, investments, household consumption, and government consumption, evaluated at horizons of 1, 2, 4, and 8 quarters ahead.

Tables from 4 to 7 report the percentage change in RMSE relative to the AR(1) benchmark model. A positive (negative) sign indicates an improvement (worsening) in forecasting performance. The forecasting performances correspond to the best-performing configurations identified through the validation procedure. Particularly, Table 4 refers to *Mod(1)*, Table 5 to *Mod(2)*, Table 6 to *Mod(3)*, and Table 7 to *Mod(4)*.

In Table 4, the inclusion of agents who implement their actions based on reinforcement learning, hereafter simply referred to as RL agents, worsens the forecasts for all variables except household consumption. The deterioration in predictive performance increases with the number of RL agents, with a particularly sharp decline in the case of inflation dynamics.

In Table 5, including RL agents leads to results that differ significantly from the previous ones. GDP forecasts show modest improvements when 25% or 50% of agents are endowed with RL. Inflation forecasts benefit consistently from the presence of RL agents, with the largest gains observed when the share of RL agents ranges between 25% and 75%. Government consumption forecasts also improve, except for very short horizons (one quarter), and a similar pattern emerges for investment. By contrast, household consumption forecasts deteriorate once RL agents are introduced. Taken together, the best-performing configuration appears to be the one with 50% of agents using RL.

The results reported in Table 6 show that the inclusion of RL agents leads to a deterioration in the forecasts for all variables (with a few exceptions, such as non-short-term household consumption), and that this deterioration increases with the share of RL agents.

Table 7 shows a nonlinear relationship between predictive performance and the share of RL agents included in the model. For GDP in the very short term (one quarter), no variations are observed, while the forecast appears to deteriorate over longer horizons. For inflation, there is a marked improvement with 50% RL agents, a more modest improvement with 25% and 75%, and a deterioration with 100%. For investment, changes in predictive accuracy are limited, but simulations with 75% and 100% RL agents appear to perform best. The forecast for household consumption seems to improve with a higher share of RL agents at the one-quarter horizon, while tending to worsen over longer horizons when the share of RL agents exceeds 25%. Government consumption forecasts display mixed results across horizons, but overall tend to improve with 50% or 75% RL agents and worsen in other cases. The overall picture thus appears highly fragmented.

Figure 7 illustrates the performance of the four models for GDP and Inflation. In the next section, we provide a deeper analysis of the key drivers behind forecast performance.

Table 4: Out-of-sample forecast performance in comparison to AR(1) model. The stars represent the significance of the Diebold-Mariano test, which compares the predictive accuracy of two forecasts. The forecast period is 2017Q2 to 2019Q4. The AR(1) model is estimated each quarter from 2017Q1 and the time series data used for estimation begins from 1996Q1. The ABM results are obtained as an average of 500 Monte Carlo simulations and are initialized in each quarter from 2017Q1 to 2019Q3. Results refer to specification *Mod(1)*.

	GDP	Inflation	Investments	Household cons	Gov cons
	Mod(1) RL 0%				
1q	-9.6 (0.63)	-125.8 (0.00***)	-3.4 (0.72)	4.2 (0.89)	-69.6 (0.06*)
2q	-2.7 (0.92)	-95.7 (0.00***)	-1.2 (0.89)	26.5 (0.23)	-76.1 (0.06*)
4q	9.3 (0.78)	-93.2 (0.05*)	-2.3 (0.27)	27.1 (0.18)	-82.8 (0.39)
8q	15.8 (0.00***)	-141.4 (0.00***)	4.1 (0.00***)	22.8 (0.00***)	-75.9 (0.00***)
	Mod(1) RL 25%				
1q	-11.3 (0.60)	-205.8 (0.00***)	-5 (0.63)	12.2 (0.72)	-125.5 (0.00***)
2q	-17 (0.65)	-148.6 (0.00***)	-5.5 (0.58)	29.5 (0.32)	-130.4 (0.01**)
4q	-16.9 (0.66)	-148.2 (0.02**)	-5.7 (0.20)	28.7 (0.14)	-129.9 (0.08*)
8q	-2.7 (0.00***)	-237.5 (0.00***)	0 (0.00***)	26.7 (0.00***)	-111.4 (0.00***)
	Mod(1) RL 50%				
1q	-14.9 (0.53)	-361.7 (0.00***)	-8.1 (0.49)	-24.4 (0.47)	-232 (0.00***)
2q	-58 (0.24)	-188.3 (0.00***)	-14.1 (0.29)	17.7 (0.65)	-179.2 (0.01***)
4q	-62.3 (0.19)	-203.7 (0.04**)	-15.8 (0.04**)	41.8 (0.30)	-185.3 (0.12)
8q	-21 (0.00***)	-394.7 (0.00***)	-5.9 (0.41)	37.2 (0.00***)	-169.7 (0.00***)
	Mod(1) RL 75%				
1q	-18.7 (0.46)	-504.3 (0.00***)	-13.3 (0.34)	-141.7 (0.01**)	-324.9 (0.00***)
2q	-107.1 (0.07*)	-297.4 (0.00***)	-29.9 (0.12)	-15.6 (0.77)	-252 (0.00***)
4q	-122 (0.00***)	-332.6 (0.01**)	-29.1 (0.06*)	42.6 (0.48)	-258.5 (0.08*)
8q	-60 (0.45)	-651 (0.00***)	-16.4 (0.00***)	52.9 (0.09*)	-222.6 (0.02**)
	Mod(1) RL 100%				
1q	-22 (0.42)	-572.4 (0.00***)	-17.4 (0.26)	-148.7 (0.00***)	-407.5 (0.00***)
2q	-134.2 (0.03**)	-383.1 (0.00***)	-38.5 (0.09*)	-38.8 (0.50)	-355.2 (0.00***)
4q	-176.1 (0.00***)	-435.6 (0.01**)	-44.7 (0.04**)	25.4 (0.70)	-358.2 (0.02**)
8q	-104.3 (0.00***)	-1073 (0.00***)	-27.9 (0.06*)	58.5 (0.00***)	-318.4 (0.00***)

ABM percentage improvements (+) or losses (-) relative to AR(1)

Table 5: Out-of-sample forecast performance in comparison to AR(1) model. The stars represent the significance of the Diebold-Mariano test, which compares the predictive accuracy of two forecasts. The forecast period is 2017Q2 to 2019Q4. The AR(1) model is estimated each quarter from 2017Q1 and the time series data used for estimation begins from 1996Q1. The ABM results are obtained as an average of 500 Monte Carlo simulations and are initialized in each quarter from 2017Q1 to 2019Q3. Results refer to specification *Mod(2)*.

	GDP	Inflation	Investments	Household cons	Gov cons
	Mod(2) RL 0%				
1q	-15.3 (0.43)	-124.7 (0.00***)	-4.4 (0.67)	-0.2 (0.99)	-77.3 (0.05*)
2q	-7.6 (0.83)	-93.9 (0.00***)	-2.2 (0.81)	26.7 (0.28)	-76.6 (0.08*)
4q	9.3 (0.63)	-94.2 (0.08*)	-2.3 (0.72)	26.8 (0.16)	-78.2 (0.47)
8q	12.5 (0.00***)	-151.6 (0.00***)	3.5 (0.64)	19.6 (0.07*)	-68.7 (0.20)
	Mod(2) RL 25%				
1q	-15.2 (0.43)	-33.2 (0.00***)	-4.7 (0.65)	7 (0.81)	-78.7 (0.06*)
2q	2.4 (0.90)	-66.1 (0.02**)	1.9 (0.80)	25.7 (0.21)	-63.4 (0.08*)
4q	12.6 (0.38)	-56 (0.10)	-1.1 (0.85)	14.8 (0.12)	-43.5 (0.62)
8q	16.6 (0.00***)	-75.3 (0.25)	3 (0.70)	14.3 (0.00***)	-40.6 (0.00***)
	Mod(2) RL 50%				
1q	-15.4 (0.42)	0.2 (0.99)	-4.7 (0.65)	13.9 (0.55)	-74.8 (0.06*)
2q	-7 (0.28)	-52.9 (0.01**)	6.2 (0.30)	17.2 (0.15)	-25 (0.50)
4q	18.6 (0.27)	-44.8 (0.08*)	1.8 (0.76)	25.4 (0.38)	-23.9 (0.80)
8q	16.2 (0.57)	-43.4 (0.00***)	4.1 (0.00***)	7.8 (0.00***)	-16.4 (0.00***)
	Mod(2) RL 75%				
1q	-16.2 (0.41)	-55.7 (0.14)	-5 (0.64)	-31.2 (0.36)	-103.3 (0.02**)
2q	-33.7 (0.16)	-6.5 (0.82)	9.6 (0.13)	1.4 (0.80)	14.2 (0.60)
4q	-29.5 (0.09*)	15.5 (0.52)	8.3 (0.27)	-30.7 (0.00***)	8.2 (0.57)
8q	-3.2 (0.00***)	-12 (0.00***)	6.5 (0.00***)	1.5 (0.00***)	65.4 (0.00***)
	Mod(2) RL 100%				
1q	-16.3 (0.42)	-117.8 (0.02**)	-5.3 (0.62)	-20.8 (0.54)	-114.7 (0.01**)
2q	-64.7 (0.06*)	-84.4 (0.09*)	9 (0.14)	-40.1 (0.03**)	-13.9 (0.73)
4q	-70 (0.01**)	5.4 (0.91)	8.2 (0.11)	-38.3 (0.13)	-41.6 (0.27)
8q	-32.9 (0.00***)	-46.4 (0.00***)	8.6 (0.00***)	0.3 (0.00***)	35.2 (0.00***)

ABM percentage improvements (+) or losses (-) relative to AR(1)

Table 6: Out-of-sample forecast performance in comparison to AR(1) model. The stars represent the significance of the Diebold-Mariano test, which compares the predictive accuracy of two forecasts. The forecast period is 2017Q2 to 2019Q4. The AR(1) model is estimated each quarter from 2017Q1 and the time series data used for estimation begins from 1996Q1. The ABM results are obtained as an average of 500 Monte Carlo simulations and are initialized in each quarter from 2017Q1 to 2019Q3. Results refer to specification *Mod(3)*.

	GDP	Inflation	Investments	Household cons	Gov cons
			Mod(3) RL 0%		
1q	-15.3 (0.43)	-124.7 (0.00***)	-4.4 (0.67)	-0.2 (0.99)	-77.3 (0.05*)
2q	-7.6 (0.83)	-93.9 (0.00***)	-2.2 (0.81)	26.7 (0.28)	-76.6 (0.08*)
4q	9.3 (0.63)	-94.2 (0.08*)	-2.3 (0.72)	27.3 (0.17)	-78.2 (0.47)
8q	8.8 (0.00***)	-150.9 (0.00***)	3.4 (0.00***)	19.3 (0.00***)	-68.5 (0.00***)
			Mod(3) RL 25%		
1q	-16.4 (0.41)	-140.9 (0.00***)	-6.1 (0.58)	-19 (0.56)	-107.6 (0.01**)
2q	-15.3 (0.71)	-97 (0.01**)	-6.4 (0.56)	26 (0.29)	-95.4 (0.05*)
4q	5 (0.88)	-92.1 (0.09*)	-3.8 (0.06*)	29.4 (0.21)	-88.5 (0.45)
8q	7.6 (0.00***)	-128.8 (0.14)	0 (0.99)	18.8 (0.08*)	-91.6 (0.12)
			Mod(3) RL 50%		
1q	-18.8 (0.37)	-215.1 (0.00***)	-8.8 (0.47)	-61.7 (0.12)	-162.3 (0.00***)
2q	-39.4 (0.46)	-123.4 (0.00***)	-13.8 (0.36)	19.8 (0.61)	-130.4 (0.05*)
4q	-23.9 (0.57)	-131.7 (0.10)	-13.2 (0.21)	34.6 (0.22)	-136.3 (0.36)
8q	-6.4 (0.92)	-241.9 (0.00***)	-5.8 (0.00***)	21.2 (0.00***)	-134.7 (0.03**)
			Mod(3) RL 75%		
1q	-20.1 (0.36)	-222.3 (0.00***)	-11.7 (0.38)	-94.5 (0.09*)	-204.7 (0.00***)
2q	-43.6 (0.38)	-130.2 (0.00***)	-16.6 (0.28)	28.5 (0.46)	-164.5 (0.04**)
4q	-16.7 (0.66)	-144.3 (0.04**)	-12.5 (0.25)	36.1 (0.24)	-159.5 (0.32)
8q	-4.8 (0.00***)	-202.8 (0.00***)	-6.1 (0.36)	19.9 (0.08*)	-150.7 (0.00***)
			Mod(3) RL 100%		
1q	-22.4 (0.34)	-281.4 (0.00***)	-16 (0.28)	-127.2 (0.02**)	-269.9 (0.00***)
2q	-66.7 (0.24)	-189.9 (0.00***)	-29 (0.17)	24.6 (0.61)	-219.4 (0.01**)
4q	-56.9 (0.14)	-203.1 (0.02**)	-25 (0.16)	41.4 (0.23)	-224.3 (0.18)
8q	-34.8 (0.00***)	-266.2 (0.00***)	-15 (0.00***)	27.1 (0.00***)	-185.8 (0.00***)

ABM percentage improvements (+) or losses (-) relative to AR(1)

Table 7: Out-of-sample forecast performance in comparison to AR(1) model. The stars represent the significance of the Diebold-Mariano test, which compares the predictive accuracy of two forecasts. The forecast period is 2017Q2 to 2019Q4. The AR(1) model is estimated each quarter from 2017Q1 and the time series data used for estimation begins from 1996Q1. The ABM results are obtained as an average of 500 Monte Carlo simulations and are initialized in each quarter from 2017Q1 to 2019Q3. Results refer to specification *Mod(4)*.

	GDP	Inflation	Investments	Household cons	Gov cons
			Mod(4) RL 0%		
1q	-9.6 (0.63)	-125.8 (0.00***)	-3.4 (0.72)	4.2 (0.89)	-69.6 (0.06*)
2q	1.1 (0.97)	-95.7 (0.00***)	-0.4 (0.96)	25.9 (0.26)	-83.4 (0.06*)
4q	23.6 (0.34)	-96.1 (0.08*)	3.1 (0.41)	27.9 (0.20)	-67.9 (0.41)
8q	13 (0.00***)	-140 (0.00***)	9.7 (0.00***)	21.8 (0.00***)	-43.7 (0.00***)
			Mod(4) RL 25%		
1q	-10.6 (0.61)	-117.2 (0.01***)	-3.5 (0.72)	7.2 (0.78)	-98.1 (0.01**)
2q	-5.6 (0.83)	-45.4 (0.01***)	-1.7 (0.84)	33.6 (0.24)	-67.7 (0.10*)
4q	-10.8 (0.71)	-27.7 (0.13)	-6.8 (0.22)	25.1 (0.31)	-78.8 (0.39)
8q	-10.9 (0.00***)	-39.1 (0.00***)	-3.9 (0.00***)	19.1 (0.10*)	-96.7 (0.00***)
			Mod(4) RL 50%		
1q	-10.5 (0.61)	-0.1 (0.99)	-3.5 (0.72)	24.4 (0.26)	-93.1 (0.02**)
2q	-6.5 (0.66)	3.5 (0.85)	6 (0.41)	2.7 (0.84)	-5.2 (0.82)
4q	24.9 (0.28)	18.6 (0.44)	1.6 (0.74)	4.6 (0.03**)	-10 (0.83)
8q	9.4 (0.00***)	1.3 (0.00***)	2 (0.00***)	17.9 (0.00***)	-29.1 (0.00***)
			Mod(4) RL 75%		
1q	-10.3 (0.62)	-108.4 (0.03**)	-3.3 (0.73)	22.6 (0.26)	-94.1 (0.02**)
2q	-59.2 (0.10)	-91.8 (0.23)	13.1 (0.19)	-74.3 (0.00***)	21.3 (0.45)
4q	-19.9 (0.52)	-24.1 (0.76)	12.3 (0.17)	-26.2 (0.28)	-5 (0.88)
8q	9.8 (0.00***)	-28.2 (0.00***)	10 (0.32)	4.3 (0.00***)	11.7 (0.00***)
			Mod(4) RL 100%		
1q	-10.6 (0.61)	-332.3 (0.00***)	-3.2 (0.74)	22.2 (0.24)	-102.3 (0.01**)
2q	-156.2 (0.01***)	-338.5 (0.00***)	19.6 (0.26)	-242.2 (0.00***)	-93.1 (0.07*)
4q	-150.4 (0.02**)	-225.7 (0.10*)	26.5 (0.31)	-124.1 (0.02**)	-107.5 (0.17)
8q	-62.1 (0.00***)	-431.8 (0.00***)	23.4 (0.00***)	-48.2 (0.00***)	-10.2 (0.00***)

ABM percentage improvements (+) or losses (-) relative to AR(1)

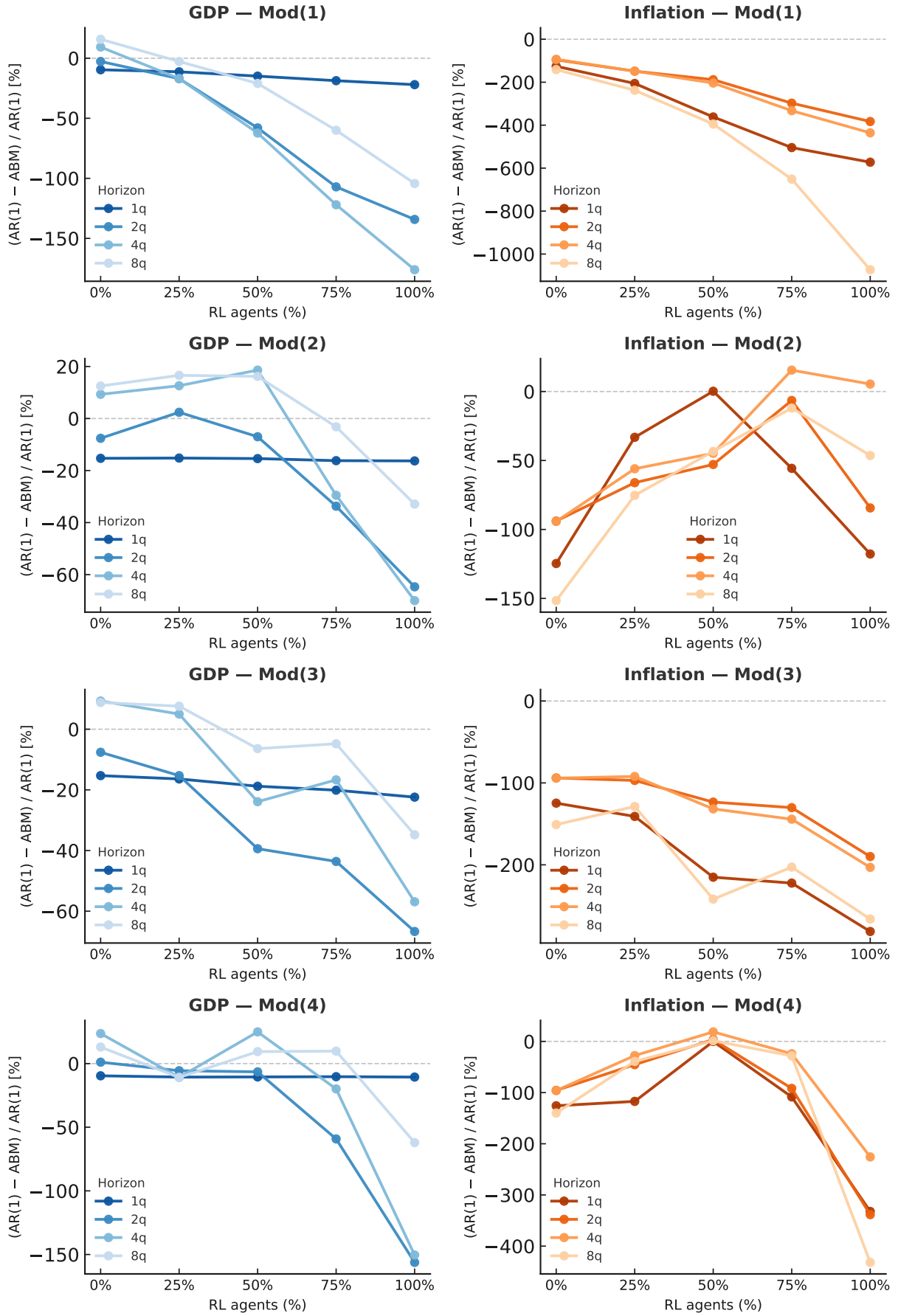


Figure 7: Out-of-sample performance $(AR(1) - ABM)/AR(1)$ % RMSE across variables and specifications Mod(1)–Mod(4). Lines indicate forecast horizons (1q, 2q, 4q, 8q). The x-axis shows the share of RL firms

5.1. Results drivers

Given the many variables involved in the training and testing phases, identifying the primary drivers of forecasting performance is not straightforward. To address this, we employ a linear regression model that uses the training hyperparameters and simulation parameters as input features and the synthetic forecast difference (average difference of RMSE between AR(1) and ABM across all macroeconomic variables and forecast horizons) as the dependent variable. In addition to the model configurations presented above, alternative combinations of forecasted values were also computed to expand the dataset of forecasts and enhance the statistical soundness of the results, that are reported in Table 8. After removing heavily correlated variables and variables with zero variance, the regression results below suggest a statistically significant model that explains 63% of the variance in forecast performance. Notably, RLF^2 (representing the squared percentage of firms performing RL during the testing phase) is statistically significant (p-value = 0.004), whereas the linear term is not. This finding points to a non-linear relationship between RLF and the dependent variable. The positive coefficient of the first order term and the negative coefficient of the squared term imply a concave effect: increasing RLF increases ABM forecast performance up to an intermediate level (23,6%, that is near 25%), beyond which further increases in RLF lead to worse ABM prediction power. In particular, with up to 47.2% of firms using RL (essentially 50%), the forecast appears to slightly improve compared to the case without the use of MARL. Above this threshold, the forecast tends to worsen more and more markedly the closer one gets to 100% of RL firms.

$\Delta\epsilon$ is the drop in ϵ between its initial and final values during training and $\Delta\phi$ is the width of the price grid that defines the agent's actions ($\phi_{max} - \phi_{min}$). The variables are both statistically significant, with coefficients of opposite sign: positive and negative, respectively. This finding implies that greater reductions in ϵ (i.e., more pronounced exploration-exploitation shifts during training) and a narrower action space are associated with improved ABM predictions. It is important to note that the action space cannot be expanded indefinitely, as larger action spaces require substantially more simulations to achieve adequate coverage. In practice, a narrower action space can lead to better performance, since broader spaces demand more extensive exploration. Lastly, α (firm size in the search-and-match process) is positively associated with prediction ability. This relationship suggests that size matters: a search-and-match mechanism based on both price and firm size leads to improved forecasting performance in the ABM.

Table 8: OLS Regression Output

Dep. Variable :	value_error	R-squared :	0.629
Model :	OLS	Adj. R-squared :	0.598
Method :	Least Squares	F-statistic :	20.04
Date :	Sat, 04 Oct 2025	Prob (F-statistic) :	1.26e-11
Time :	08:36:33	Log-Likelihood :	-338.99
No. Observations :	65	AIC :	690.0
Df Residuals :	59	BIC :	703.0
Df Model :	5		
Covariance Type :	nonrobust		

Variable	Coef.	Std. Err.	t	P> t	[0.025	0.975]
const	-86.7405	8.070	-10.749	0.000	-102.888	-70.593
$\Delta\Phi$	-18.5700	6.976	-2.662	0.010	-32.528	-4.612
$\Delta\epsilon$	46.5888	7.141	6.524	0.000	32.300	60.878
RLF	29.3749	20.600	1.426	0.159	-11.845	70.595
RLF^2	-62.2629	20.600	-3.022	0.004	-103.483	-21.043
α	30.3515	12.161	2.496	0.015	6.016	54.687

6. Conclusion

This paper addresses a common critique of traditional agent-based models (ABMs), namely their reliance on simplistic heuristics for agents' behavior, which are therefore modeled in an overly irrational way and with an excess of degrees of freedom due to the wide range of possible behavioral rules. We thus introduce a reinforcement learning mechanism to enhance firms' pricing strategies within a large, data-driven ABM (the model of Poledna et al., 2023), calibrated to the Italian economy (Di Domenico et al., 2025). Starting from an empirically grounded representation of the Italian firm population, albeit scaled, the model enables the evaluation of firms' optimal pricing strategies based on their structural characteristics, the sector in which they operate, and their relative market share, within an environment that closely mirrors the actual macroeconomic dynamics of the Italian economy.

We find that firms trained with a reinforcement learning mechanism exhibit mean-reverting price-setting behavior, adjusting prices in response to deviations from the sectoral average, which aligns with empirical observations. In other words, price adjustment dynamics tend to converge toward the market average. This finding reflects how individual pricing strategies are shaped by competitive pressures and market feedback, ultimately leading to price coordination across firms. This mechanism is already present in other ABMs that implement heuristics to represent such behavior. Moreover, the resulting dynamics are more complex when firm size affects the search and matching process. Smaller firms, especially in densely populated sectors, tend to cut prices more aggressively to protect or expand their market share, which slightly lowers the mean-reverting pattern. By contrast, larger firms operating in less competitive sectors display more heterogeneous pricing behavior, indicating that some take advantage of their market power and deviate from the mean-reverting tendency.

Subsequently, we apply the pricing decisions generated by reinforcement learning to varying shares of firms and examine how the out-of-sample forecast fit of several macroeconomic variables, such as GDP and inflation, is affected. In some settings, we find that the inclusion of firms using reinforcement learning worsens the forecasts, while in others we observe that even a substantial share (e.g., 50%) of RL firms improves them.

We therefore conclude that, on the one hand, using sensible heuristics, preferably grounded in empirical observations (such as those derived from behavioral economics studies), may be sufficient to achieve a well-performing model, and certainly better than models in which all agents are assumed to use RL. On the other hand, considering that a non-negligible minority of firms (i.e., 25%-50%) base their decisions on mechanisms more complex than too simple heuristics can be beneficial for the predictive performance of the macroeconomic model, and also for mitigating the problem of the “wilderness” of behavioral functions. Regarding this latter aspect, a heuristic rule, which may appear arbitrary, is replaced with a rule derived from a reinforcement learning mechanism based on a clear and commonly accepted reward, namely profit maximization. It is true, however, that the use of MARL still entails arbitrary choices concerning the modeled economic mechanisms and the hyperparameters of the MARL methodology.

Appendix A. Symbols and notation

Table A.9: Summary of symbols and notation

Symbol	Description
Households and consumption	
$Y_h^e(t)$	Expected disposable income of household h .
$w_h(t)$	Wage of worker h .
$sb_{other}(t)$	Additional social transfers.
$C_h^d(t)$	Consumption of household h .
$\bar{P}^{HH}(t)$	Consumer price index.
ψ	Marginal propensity to consume.
$\pi^e(t)$	Expected inflation.
θ^{UB}	Replacement rate of unemployment benefits.
θ^{DIV}	Share of profits distributed as dividends.
τ^{SIW}	Social security contribution rate on workers.
τ^{INC}	Income tax rate.
τ^{VAT}	Value-added tax rate.
Firms and production	
$Y_i(t)$	Output of firm i .
$N_i(t), M_i(t), K_i(t)$	Labour, intermediates, and capital of firm i .
$I_i^d(t)$	Desired investment of firm i .
$K_i(t)$	Capital stock of firm i .
$\Pi_i(t)$	Gross profit of firm i .
$\pi_{i,t}$	Profit rate of firm i .
$\pi_{i,t}^s$	Average sectoral profit rate.
$p_i(t)$	Price set by firm i .
$\pi_i(t)$	Cost-push inflation of firm i .
$\mu_i(t)$	Price adjustment chosen via RL.
$\alpha_i, \beta_i, \kappa_i$	Productivity parameters (labour, intermediates, capital).
δ_i	Depreciation rate of capital.
S_i	Firm size used in matching (total supply).
Government and fiscal policy	
$\Delta\%CG(t)$	Growth rate of government consumption.
α_G, β_G	AR(1) and trend parameters of government spending.
$\Pi_G(t)$	Government budget balance.
$L_G(t)$	Public debt.
τ^{FIRM}	Corporate income tax rate.
External sector	
$\Delta\%Y_I(t)$	Growth rate of imports.
$\Delta\%CE(t)$	Growth rate of exports.
α_I, α_E	Persistence of import/export.
β_I, β_E	Trend of import/export.

Table A.9 (continued)

Symbol	Description
ϵ_I, ϵ_E	Foreign stochastic shocks.
Monetary policy	
$\bar{r}(t)$	Policy interest rate (Taylor rule).
ρ	Interest-rate smoothing parameter.
r^*	Target real interest rate.
π^*	Inflation target.
ξ_π, ξ_γ	Central bank responsiveness.
$\pi^{EA}(t), \gamma^{EA}(t)$	Euro area inflation and growth.
Matching (market clearing)	
π_i^{price}	Price-based matching probability.
π_i^{size}	Size-based matching probability.
$\pi_i^{comb}(\alpha)$	Combined matching probability.
α	Weight of size in the combined probability.
RL: states and actions	
$s_{i,t}$	State of firm i at time t .
$a_{i,t}$	Action chosen by firm i at time t .
$\phi_{\min}, \phi_{\max}$	Min/max price adjustment factors.
Δp	Price-change bin width for discretization.
$\mu_i(t)$	Price adjustment actually applied by firm i .
RL: Q-learning and Bellman	
$Q_i(s_{i,t}, a_{i,t})$	Q-table of firm i .
α_Q	Learning rate.
γ_Q	Discount factor.
ϵ_Q	Exploration rate in ϵ -greedy policy.
RL: reward	
$r_{i,t}$	Reward of firm i at time t .
$\pi_{i,t}$	Firm i 's profit rate used in the reward.
$\bar{\pi}_{s(i),t}$	Sectoral mean profit rate.
Training, testing, stability	
$M \in \{\text{my}, \text{myp}\}$	Model specification: linear (my) or penalised (myp) reward.
E_{train}, E_{test}	Number of training and testing epochs.
T	Periods per epoch.
OFS	Share of rational firms in testing.
$S I_t(s)$	Stability Index for state s .
$S I$	Stability Index (0–1).

Appendix A. Additional Data

Table A.10: Combinations of Agents (IDs 40-45)

Group Size	Combinations	Total
1	(40), (41), (42), (43), (44), (45)	6
2	(40, 41), (40, 42), (40, 43), (40, 44), (40, 45), (41, 42), (41, 43), (41, 44), (41, 45), (42, 43), (42, 44), (42, 45), (43, 44), (43, 45), (44, 45)	15
3	(40, 41, 42), (40, 41, 43), (40, 41, 44), (40, 41, 45), (40, 42, 43), (40, 42, 44), (40, 42, 45), (40, 43, 44), (40, 43, 45), (40, 44, 45), (41, 42, 43), (41, 42, 44), (41, 42, 45), (41, 43, 44), (41, 43, 45), (41, 44, 45), (42, 43, 44), (42, 43, 45), (42, 44, 45), (43, 44, 45)	20
4	(40, 41, 42, 43), (40, 41, 42, 44), (40, 41, 42, 45), (40, 41, 43, 44), (40, 41, 43, 45), (40, 41, 44, 45), (40, 42, 43, 44), (40, 42, 43, 45), (40, 42, 44, 45), (40, 43, 44, 45), (41, 42, 43, 44), (41, 42, 43, 45), (41, 42, 44, 45), (41, 43, 44, 45), (42, 43, 44, 45)	15
5	(40, 41, 42, 43, 44), (40, 41, 42, 43, 45), (40, 41, 42, 44, 45), (40, 41, 43, 44, 45), (40, 42, 43, 44, 45), (41, 42, 43, 44, 45)	6
6	(40, 41, 42, 43, 44, 45)	1

Table A.11: Number of firms and employees per sector, at different scale factors. The first column identifies the different sectors corresponding to the subsequent columns. Period 2015 Q1. An overview of the sectors analyzed in our study, including their unique Sector_IDs is provided in Table A.12

Firms (Total: 4,823,672 for Scale 1 to 1 - 2015 Q1)																
1-16	389063	5849	8540	2209	57489	60304	28511	12866	6256	284	3379	1439	10119	19444	25278	42529
17-32	5137	8358	23027	3305	1455	48229	39624	10856	1441	7900	520915	116726	390601	622931	98039	1711
33-48	223	23059	2551	326383	5071	8627	4439	82574	3850	217	88087	239911	343248	203174	8919	20377
49-62	145821	13719	1097	16377	111870	10000	30077	272348	15284	36185	29074	30508	21568	155220		
Employees (Total: 17,401,829 for Scale 1 to 1 - 2015 Q1)																
1-16	769030	14143	25923	31974	429938	463404	107411	88541	62549	11100	101095	62432	171013	159012	150320	467197
17-32	105412	142080	451674	154138	88075	251660	161456	89174	28152	159048	1329421	360887	1125483	1832040	518747	48580
33-48	20043	343862	159897	1330782	33463	50416	84712	375221	361074	45706	164326	299878	629165	287452	24788	66619
49-62	208549	42674	257373	46458	821895	251139	97097	513849	313735	75386	79295	112631	41142	302093		
Firms (Total: 4,824 for Scale 1 to 1000 - 2015 Q1)																
1-16	389	5	8	2	57	60	28	12	6	1	3	1	10	19	25	42
17-32	5	8	23	3	1	48	39	10	1	7	520	116	390	622	98	1
33-48	1	23	2	326	5	8	4	82	3	1	88	239	343	203	8	20
49-62	145	13	1	16	111	10	30	272	15	36	29	30	21	155		
Employees (Total: 17,402 for Scale 1 to 1000 - 2015 Q1)																
1-16	769	14	25	32	429	463	107	88	62	11	101	62	171	159	150	467
17-32	105	142	451	154	88	251	161	89	28	159	1329	360	1125	1832	518	48
33-48	20	343	159	1330	33	50	84	375	361	45	164	299	629	287	24	66
49-62	208	42	257	46	821	251	97	513	313	75	79	112	41	302		
Firms (Total: 82 for Scale 1 to 100,000 - 2015 Q1)																
1-16	3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
17-32	1	1	1	1	1	1	1	1	1	1	5	1	3	6	1	1
33-48	1	1	1	3	1	1	1	1	1	1	1	2	3	2	1	1
49-62	1	1	1	1	1	1	1	2	1	1	1	1	1	1		
Employees (Total: 169 for Scale 1 to 100,000 - 2015 Q1)																
1-16	7	1	1	1	4	4	1	1	1	1	1	1	1	1	1	4
17-32	1	1	4	1	1	2	1	1	1	1	13	3	11	18	5	1
33-48	1	3	1	13	1	1	1	3	3	1	1	2	6	2	1	1
49-62	2	1	2	1	8	2	1	5	3	1	1	1	1	3		

Table A.12: List of the 62 sectors

No.	Sector_ID	Description
1	A01	Products of agriculture, hunting and related services
2	A02	Products of forestry, logging and related services
3	A03	Fish and other fishing products; aquaculture products; support services to fishing
4	B	Mining and quarrying
5	C10-12	Food, beverages and tobacco products
6	C13-15	Textiles, wearing apparel, leather and related products
7	C16	Wood and of products of wood and cork, except furniture; articles of straw and plaiting materials
8	C17	Paper and paper products
9	C18	Printing and recording services
10	C19	Coke and refined petroleum products
11	C20	Chemicals and chemical products
12	C21	Basic pharmaceutical products and pharmaceutical preparations
13	C22	Rubber and plastic products
14	C23	Other non-metallic mineral products
15	C24	Basic metals
16	C25	Fabricated metal products, except machinery and equipment
17	C26	Computer, electronic and optical products
18	C27	Electrical equipment
19	C28	Machinery and equipment n.e.c.
20	C29	Motor vehicles, trailers and semi-trailers
21	C30	Other transport equipment
22	C31-32	Furniture and other manufactured goods
23	C33	Repair and installation services of machinery and equipment
24	D	Electricity, gas, steam and air conditioning
25	E36	Natural water; water treatment and supply services
26	E37-39	Sewerage services; sewage sludge; waste collection, treatment and disposal services; materials recovery services; remediation services and other
27	F	Constructions and construction works
28	G45	Wholesale and retail trade and repair services of motor vehicles and motorcycles
29	G46	Wholesale trade services, except of motor vehicles and motorcycles
30	G47	Retail trade services, except of motor vehicles and motorcycles
31	H49	Land transport services and transport services via pipelines
32	H50	Water transport services
33	H51	Air transport services
34	H52	Warehousing and support services for transportation
35	H53	Postal and courier services
36	I	Accommodation and food services
37	J58	Publishing services
38	J59-60	Motion picture, video and television programme production services, sound recording and music publishing; programming and broadcasting services
39	J61	Telecommunications services
40	J62-63	Computer programming, consultancy and related services; Information services
41	K64	Financial services, except insurance and pension funding
42	K65	Insurance, reinsurance and pension funding services, except compulsory social security
43	K66	Services auxiliary to financial services and insurance services
44	L68A	Real estate services
45	M69-70	Legal and accounting services; services of head offices; management consultancy services
46	M71	Architectural and engineering services; technical testing and analysis services
47	M72	Scientific research and development services
48	M73	Advertising and market research services
49	M74-75	Other professional, scientific and technical services and veterinary services
50	N77	Rental and leasing services
51	N78	Employment services
52	N79	Travel agency, tour operator and other reservation services and related services
53	N80-82	Security and investigation services; services to buildings and landscape; office administrative, office support and other business support services
54	O	Public administration and defence services; compulsory social security services
55	P	Education services
56	Q86	Human health services
57	Q87-88	Residential care services; social work services without accommodation
58	R90-92	Creative, arts, entertainment, library, archive, museum, other cultural services; gambling and betting services
59	R93	Sporting services and amusement and recreation services
60	S94	Services furnished by membership organisations
61	S95	Repair services of computers and personal and household goods
62	S96	Other personal services

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